Ore Geology Reviews 76 (2016) 52–78

Contents lists available at ScienceDirect

Ore Geology Reviews

journal homepage: www.elsevier.com/locate/oregeorev

Review

Gallium, germanium, indium, and other trace and minor elements in sphalerite as a function of deposit type — A meta-analysis

Max Frenzel [a,*], Tamino Hirsch [a], Jens Gutzmer [a,b]

[a] Helmholtz-Zentrum Dresden-Rossendorf, Helmholtz Institute Freiberg for Resource Technology, Halsbrücker Str. 34, 09599 Freiberg, Germany
[b] Institute for Mineralogy, University of Technology Bergakademie Freiberg, Brennhausgasse 14, 09599 Freiberg, Germany

ARTICLE INFO

Article history:
Received 24 June 2015
Received in revised form 6 December 2015
Accepted 21 December 2015
Available online 14 January 2016

Keywords:
High-tech metals
By-products
Fluid salinity
Sphalerite geothermometer
Fluid inclusions
Homogenisation temperature
GGIMFis thermometer

ABSTRACT

While a significant amount of analytical data on trace and minor element concentrations in sphalerite has been collected over the last six decades, no meta-analysis of this data has ever been conducted. In this study, the results of such an analysis are presented. While the study focusses on Ga, Ge and In, data for six other elements (Ag, Cd, Co, Cu, Fe and Mn) was also included.

The results show that there are systematic, statistically significant differences in the mean concentrations of Fe, Ga, Ge, In and Mn in sphalerite from different deposit types, while Cd and Cu concentrations show no such differences, and Ag and Co concentrations are only significantly different for vein-type deposits. A principal component analysis demonstrates that the differences between deposit types are approximately one-dimensional, being expressible in terms of a single number. This number correlates strongly with the homogenisation temperature of fluid inclusions ($R^2 = 0.82$, $p < 2\times10^{-16}$). It may be expressed as follows:

$$\mathrm{PC\ 1}^* = \ln\left(\frac{C_{\mathrm{Ga}}^{0.22}\cdot C_{\mathrm{Ge}}^{0.22}}{C_{\mathrm{Fe}}^{0.37}\cdot C_{\mathrm{Mn}}^{0.20}\cdot C_{\mathrm{In}}^{0.11}}\right)$$

with Ga, Ge, In and Mn concentrations in ppm, and Fe concentration in wt.%. The relationship is sufficiently strong to be used as a geothermometer (GGIMFis). The empirical relationship between PC 1* and the homogenisation temperature, *T*, is:

$$T(^\circ\mathrm{C}) = (54.4 \pm 7.3)\cdot \mathrm{PC\ 1}^* + (208 \pm 10).$$

Our results indicate a strong control of sphalerite chemistry by fluid temperature, particularly for the concentrations of Ga ($R^2 \sim 0.40$), Ge ($R^2 \sim 0.65$), Fe ($R^2 \sim 0.30$) and Mn ($R^2 \sim 0.60$), and to a lesser degree In ($R^2 \sim 0.10$). The concentrations of Ag, Cd, Co and Cu appear to be independent of temperature.

As a consequence of the strong temperature control on PC 1*, metamorphic overprinting of Pb–Zn deposits, even by lower greenschist facies events, may lead to significant changes in sphalerite composition, namely a relative decrease in Ga and Ge concentrations, and increase in Fe, Mn and, to a lesser degree, In concentrations. The closure temperature of sphalerite in regional metamorphic events appears to be around 310 ± 50 °C, such that higher-grade events will not be reflected in its composition.

Factors other than temperature, such as differences in fluid salinity or source-rock composition, do not appear to be responsible for differences between deposit types, but rather appear to cause differences between individual deposits. Particularly, the Cu activity in ore-forming systems appears to have a strong influence on In concentrations in sphalerite.

The observed trends in sphalerite compositions provide a useful tool for future studies of different types of Pb–Zn deposits, as well as for mineral exploration. They should be particularly relevant for the identification of new resources of Ga, Ge and In.

* Corresponding author.
E-mail address: m.frenzel@hzdr.de (M. Frenzel).

http://dx.doi.org/10.1016/j.oregeorev.2015.12.017

Contents

1. Introduction

Of the many trace elements commonly found in sphalerite, gallium (Ga), germanium (Ge) and indium (In) are currently the most relevant. This is due not only to their interesting geological behaviour (e.g. Cook et al., 2009), but also to their growing usage in certain high-tech applications such as smartphones (Ga, In), fibre-optic cables (Ge) and solar cells (Ga, Ge, In) (Guberman, 2015; Jaskula, 2015; Tolcin, 2015). The rapid increase in their economic importance as well as supply security-related concerns are reflected in their recent identification as critical raw materials (Erdmann and Graedel, 2011; EU Commission, 2014). Sulphidic zinc ores constitute a major source of all three elements (Guberman, 2015; Jaskula, 2015; Tolcin, 2015) and sphalerite is the principal host mineral in these ores (Cook and Ciobanu, 2015; George et al., 2015a,b). Yet despite the increasing economic importance of all three elements, relatively little is known about the factors controlling their occurrence in sphalerite.

All three elements were discovered in the second half of the 19th century (Lecoq de Boisbeaudran, 1875; Reich and Richter, 1863; Winkler, 1886), two of them as impurities in sphalerite (Ga: Lecoq de Boisbeaudran, 1875; and In: Reich and Richter, 1863). However, it was not until the 1920s and 30s that the advent of arc atomic emission spectroscopy (AES) (e.g. Goldschmidt, 1930; Papish et al., 1927) first allowed more detailed studies of sphalerite chemistry to be conducted (e.g. Graton and Harcourt, 1935; Kullerud, 1953; Stoiber, 1940; Oftedal, 1941; Warren and Thompson, 1945). A comprehensive review of these early studies is given by Fleischer (1955).

Despite the fact that analytical results were not at first reliably quantifiable, and were therefore usually reported in a qualitative manner (e.g. as orders of magnitude), some of the early workers (e.g. Oftedal, 1941; Warren and Thompson, 1945) suggested the presence of certain systematic trends. Namely, they described an association of high Ga and Ge, and low Fe, In and Mn concentrations to 'low-temperature' deposits, and an association of low Ga and Ge, and high Fe, In and Mn concentrations to 'high temperature' deposits. This suggestion was based on general geological criteria and was never substantiated by relevant thermometric data or appropriate statistical analyses.

Over time, the quality of AES data improved, and from the mid-1950s onwards might be considered sufficient to provide quantitative information. Unfortunately, this improvement in data quality was not accompanied by an improvement in data treatment, and suggestions similar to the one referred to above were made by the authors of the larger studies published at this time (El Shazly et al., 1957; Schroll, 1954, 1955). That is, the trace element content of sphalerite was still cited as being related to the temperature of ore formation, while a statistical analysis as well as additional data directly supporting this claim were never provided. The first authors to criticise such a simplistic treatment were Burnham (1959) and Rose (1967) who recognised that other geological factors like fluid sources and pathways might also play an important role in determining the geochemical signature of sphalerite via their influence on fluid chemistry. A number of studies followed throughout the 1960s, 70s and 80s (e.g. Chakrabarti, 1967; Hall and Heyl, 1968; Jolly and Heyl, 1968; Olade and Morton, 1985; So, 1977; Watling, 1976), but did not add substantially to the understanding of global trends.

The first study to attempt a discrimination of sphalerite samples from different deposit types based on their trace element content was published by Qian (1987). While it indicated that Fe concentration and the Ga/In ratio might be useful discriminators, it also suffers from a lack of information about analytical procedures and geological background, as well as a detailed explanation of the non-standard deposit classification used. The extensive datasets produced by the previous workers cited above were not considered and its usefulness is therefore limited.

In the same year, Möller (1987) proposed that Ga/Ge ratios in sphalerite could be used as a geothermometer. However, his treatment assumed that the Ga/Ge ratio in sphalerite and coexisting hydrothermal fluid would be identical, an assumption that seems implausible given the strong dependence of partitioning coefficients on various physicochemical parameters, including temperature itself (cf. McIntire, 1963).

Neither this assumption nor the utility of the thermometer itself were ever subjected to rigorous testing. The chemical and thermometric data plotted by Möller (1987) showed a clear deviation from the predicted trend, and most of the original thermometric data itself was not provided in the publication.

Only recently have new studies been undertaken in an attempt to assess differences in the trace element signature of sphalerite from different types of Pb–Zn deposits (Cook et al., 2009; Ye et al., 2011; Belissont et al., 2014). While the authors of these studies noted that certain petrogenetic factors such as metal source and deposit type appear to have a systematic influence on the trace element signature of sphalerite (Cook et al., 2009), they also stated that their studies lack the statistical power to reliably distinguish the characteristics of different deposit types: 'Concentrations of a given element vary significantly between samples and even grains of sphalerite in the same sample. More data are required to support some of the trends identified here and to correlate them with the rules governing sphalerite trace element geochemistry' (Cook et al., 2009).

To our best knowledge, no attempt has yet been made of a meta-analysis of all (or at least the majority of) suitable analytical data published since the mid-1950s. Such an analysis would have the advantage of being able to overcome the difficulties associated with the limited size of previous studies and the concomitant lack of statistical power. It should therefore be in a better position to identify the most relevant global trends. The objective of this work is to present the results of such a meta-analysis, using suitable statistical techniques to test for significant differences in the trace element contents of sphalerite from a number of broadly defined types of Pb–Zn deposits. Based on the results of a principal component analysis, data on fluid temperatures and salinities, as well as a detailed consideration of the physico-chemical factors affecting the partitioning behaviour of different elements between sphalerite and fluid, an explanation is proposed for the geological factor(s) causing the differences identified in the statistical analysis. This is followed by an evaluation of the implications of this work for the identification of new potential sources for the high-tech metals Ga, Ge and In.

2. Materials and methods

2.1. Chemical data

A collection of data on Ga and Ge concentrations in sphalerite (Frenzel et al., 2014, 2016) served as the basis for the database used in this study. Although the original database was expanded with data on a large number of trace and minor elements (including, wherever available: Ag, As, Bi, Cd, Co, Cu, Fe, Ga, Ge, Hg, In, Mn, Ni, Pb, Sn, Sb, Tl), the focus was on Ga, Ge and In. That is, studies without data on at least one of these three elements were generally excluded. Because different analytical techniques were used for the collection of the primary data, quality filters had to be applied to ensure approximate homogeneity of the final dataset. In particular, no data points were included for which the detection limits of Ga, Ge or In exceeded 20 ppm, or where the relative (internal) error of measurement was >50%. This resulted in the complete exclusion of data acquired by electron-probe microanalysis. In the final dataset, the detection limits for Ga, Ge and In, as well as many of the other elements, are generally below 5 ppm. Bias towards smaller deposits with anomalously high concentrations of certain elements was avoided mostly via the exclusion of studies focussing exclusively on such deposits (e.g. Burke and Kieft, 1980; Cook et al., 2011). A detailed list of the references included in the final dataset is provided in Table 1. Most of the relevant data was compiled from publications of the 1950s to 1970s, and therefore measured by AES on bulk samples. Potential issues arising from the nature of this data (mineral inclusions, different analytical methods) are discussed in more detail in Section 2.4 below.

2.2. Fluid temperature and salinity data

Fluid temperature and salinity data was compiled for 51 of the more than 500 deposits included in the chemical database. The most important constraint on the size of this dataset was the availability of suitable thermometric data: Due to the challenges involved in fluid inclusion work (e.g. Yardley and Bodnar, 2014), individual studies tend to focus on this aspect and are often restricted to single deposits. Studies of the trace element chemistry of sphalerite generally cover several deposits (cf. Table 1), but tend to have a similarly narrow focus: none of the studies included in the chemical database also contained thermometric data. Therefore, the set of deposits for which both types of data are available is quite restricted.

The separate collection of thermometric and chemical data also means that sample sets are generally not identical. That is, the thermometric data was generally collected on a different set of samples than the chemical data for each deposit. While this represents a problem for the reconciliation of the two types of data on the deposit scale, it does not represent a major issue for the global study attempted in this article. The main requirement for our purposes is that estimated mean temperatures, salinities and sphalerite compositions for each deposit, as well as their respective uncertainties, are well constrained. To ensure this, a number of selection criteria were applied to the deposits included in the dataset. Namely, strong preference was given to deposits from which chemical data was available for at least two sphalerite samples. This was done to enable the estimation of uncertainties regarding the mean composition. Furthermore, only those deposits where a minimum set of elements was analysed were included to reduce imputation errors (cf. Section 2.8). This minimum set was determined *after* the statistical analysis of the chemical data. Therefore, the details are described in Section 3.4 below. A summary of the thermometric and salinity data is provided in Table 2. For a detailed explanation of the deposit-type classification used in this table, please refer to Section 2.3 and Table 3 below. As a last note, it should be mentioned that the vast majority of temperatures cited in Table 2 are homogenisation temperatures of fluid inclusions.

The complete database for this article, containing both the chemical and fluid data, is available as online Supplementary material (Electronic Annex A). It includes all relevant analytical, geographic and geological information (detection limits, relative analytical errors, geographic location etc.) for each data point. Because detailed geological information on individual localities is generally not provided in the publications with the chemical data, additional references had to be included in the database. An explanation of the database structure as well as a list of the 338 references cited therein is provided in Electronic Annex B.

2.3. Deposit classification

For statistical analysis, localities were assigned to one of five major types of Pb–Zn deposits. The classification used for this purpose closely followed that used in earlier articles (Frenzel et al., 2014, 2015, 2016) and is summarised in Table 3. Three quarters of all localities could be assigned to one of the five deposits types, while for the remainder the available information did not allow for an unambiguous classification. Unclassified deposits were not included in the analysis.

2.4. Selection of elements for statistical analysis

Of the 17 elements included in the database only a subset was selected for statistical analysis, for two reasons: first, because not all elements were measured in the majority of cases (cf. Table 1), and second, because data quality was not always sufficient to warrant further analysis. The following criteria were applied: for an element to be included in the statistical analysis, it needed to be 1) measured for more than half of all data points, and 2) be present at concentrations above the respective detection limit in more than half of all cases. These criteria led to the

Table 1
Summary of references for analytical data.

No. of deposits	No. of samples	Region/district/deposit	Elements analysed	Analytical technique(s)[a]	Reference
145	223	Western USA and northern Mexico	Ag, As, Bi, Cd, Co, Ga, Ge, In, Mn, Mo, Ni, Sb, Sn, Tl	AES	Burnham (1959)
77	>77	World	Cd, Ga, Ge, In, Tl	–[b]	Feiser (1966)
45	60	World	Ag, Cd, Co, Cu, Ga, Ge, In, Mn, Pb, Sb, Sn	AES	El Shazly et al. (1957)
38	>72	China, mostly	Cd, Fe, Ga, In, Mn, Se, Te	–[b]	Qian (1987)
34	37	Alcudia Valley, Spain	Ag, As, Cd, Co, Cu, Fe, Ga, Ge, In, Mn, Sb, Sn	WD-XRF	Palero-Fernández and Martín-Izard (2005)
26	37	World	Ag, As, Bi, Cd, Co, Cu, Fe, Ga, Ge, In, Mn, Mo, Ni, Pb, Sb, Sn, Tl	LA-ICP-MS	Cook et al. (2009)
24	27	Ozark region, USA	Ag, Cd, Co, Cu, Fe, Ga, Ge, Mn, Ni	AES	Viets et al. (1992)
20	20	Central Kentucky, Tennessee and Appalachian zinc districts	Ag, Cd, Co, Cu, Fe, Ga, Ge, In, Mn, Mo, Ni, Pb, Sn	AES	Jolly and Heyl (1968)
19	187	Central district, New Mexico and Bingham district, Utah	Ag, As, Bi, Cd, Co, Fe, Ga, Ge, In, Mn, Mo, Ni, Pb, Sb, Sn	AES	Rose (1967)
17	81	Austria	Ag, As, Bi, Cd, Co, Fe, Ga, Ge, In, Mn, Mo, Ni, Pb, Sn, Tl	WD-XRF, OES, AAS	Cerny and Schroll (1995)
16	30	Korea	Ag, As, Bi, Cd, Co, Cu, Fe, Ga, Ge, Hg, In, Mn, Mo, Ni, Sb, Sn	AES	So (1977)
13	24	Illinois–Kentucky fluorite and Upper Mississippi Valley zinc–lead districts	Ag, Cd, Co, Cu, Fe, Ga, Ge, In, Mn, Ni	AES	Hall and Heyl (1968)
11	59	Coeur d'Alene district, Idaho	Cd, Co, Cu, Fe, Ga, Ge, In, Mn, Pb	AES	Fryklund and Fletcher (1956)
9	26	South China	Ag, As, Bi, Cd, Co, Cu, Fe, Ga, Ge, In, Mn, Ni, Pb, Sb, Sn, Tl	LA-ICP-MS	Ye et al. (2011)
9	12	UK	Ge	AES	Brewer et al. (1955)
9	9	Japan, Bolivia, China	Ag, Bi, Cd, Cu, Fe, Ga, In, Mn, Pb, Sb, Sn	LA-ICP-MS	Murakami and Ishihara (2013)
8	12	World	In, Sn	TIMS	Yi et al. (1995)
6	19	Norway, Australia	Ag, Bi, Cd, Co, Cu, Fe, Ga, Hg, In, Mn, Pb, Sb, Sn, Tl	LA-ICP-MS	Lockington et al. (2014)
6	>6	Eastern Australia	Ag, As, Cd, Cu, Fe, Ga, In, Mo, Ni, Pb, Sb, Sn	PIXE	Huston et al. (1995)
5	16	Griqualand West, South Africa	Ag, Cd, Co, Cu, Fe, Ga, Ge, Mn, Ni, Pb	ICP-MS	Schaefer et al. (2004)
5	5	Tyrolia, Austria	Ag, As, Bi, Cd, Co, Cu, Fe, Ga, Ge, Hg, In, Mn, Ni, Sb, Sn, Tl	AAS	Vavtar (1988)
5	5	Europe	Cd, Cu, Fe, Mn, Pb, Sn	LA-ICP-OES	Moenke-Blankenburg et al. (1994)
4	27	Canada	Ag, Cd, Fe, In	PIXE	Cabri et al. (1985)
4	13	Southern Benue Valley, Nigeria	Ag, Cd, Cu, Fe, Ge	AES	Olade and Morton (1985)
3	>10	Noailhac–Saint Salvy, France	Ag, As, Cd, Cu, Fe, Ga, Ge, In, Sb, Sn	ICP-MS, LA-ICP-MS	Belissont et al. (2014)
3	3	UK	Ga	AES	Morris and Brewer (1954)
2	13	Arlberg region, Austria	Ag, As, Bi, Cd, Co, Cu, Fe, Ga, Ge, Hg, In, Mn, Pb, Sb, Tl	AAS	Haditsch and Krainer (1991)
2	5	Gilman district, Colorado	Ag, Bi, Cd, Ga, In, Mn, Ni, Pb, Sn	AES	Lovering et al. (1978)
1	105	Gorno, Italy	Ag, Cd, Ga, Ge, Hg, Mn, Sb	AES	Fruth and Maucher (1966)
1	92	Keel Prospect, Ireland	Ag, Cd, Co, Cu, Fe, Ga, Ge, In, Mn, Ni, Pb, Sb	AES	Watling (1976)
1	78	Balmat, New York	Ag, As, Bi, Cd, Co, Cu, Ga, Ge, In, Mn, Mo, Ni, Pb, Sb, Sn	AES	Doe (1960)
1	>28	Grund mine, Germany	Cd, Fe, Ga, Ge, Hg, In, Mn, Sn, Tl	–[b]	Sperling et al. (1973)
1	27	Red-Dog mine, Alaska	Ag, As, Cd, Co, Cu, Fe, Ge, Hg, Mn, Pb, Sb, Tl	LA-ICP-MS	Kelley et al. (2004)
1	26	Ramsbeck region, Germany	Cd, Co, Fe, Ga, Ge, Hg, In, Ni, Sn	AES	Bauer et al. (1979)
1	22	Huize, China	Ag, Cd, Ga, Ge, In, Tl	LA-ICP-MS	Li et al. (2011)
1	12	Binna Valley, Switzerland	Ag, Cd, Cu, Fe, Ga, Ge, In, Mn, Sn	AES	Graeser (1969)
1	11	Tianqiao, Guizhou province, China	As, Cd, Co, Cu, Ga, Ge, In, Mo, Ni, Sb, Sn, Tl	ICP-MS	Zhou et al. (2011)
1	11	Zawar, Rajasthan, India	Ag, Cd, Co, Ga, Ge, In, Mn, Ni, Sb	AES	Chakrabarti (1967)
1	10	Fankou, China	Ag, Cd, Fe, Ga, Ge, In, Tl	AES	Xuexin (1984)
1	7	Picher ore-field, Oklahoma and Kansas	Ag, As, Bi, Cd, Co, Cu, Fe, Ga, Ge, Hg, In, Mn, Mo, Ni, Pb, Sb, Sn, Tl	AES	McKnight and Fisher (1970)
1	5	Zinkgruvan, Sweden	Ag, Cd, Co, Cu, Fe, Mn	LA-ICP-MS	Axelsson and Rodushkin (2001)
1	4	Caijiaping, South China	Bi, Cd, Co, Ga, In, Ni, Pb, Sb, Tl	ICP-MS	Dai et al. (2014)
1	>1	Serguza, Iraq	Cd, Cu, Fe, Ga, Ge, Mn, Sb	AAS	Al-Bassam et al. (1982)
1	1	Um Gheig, Egypt	Ag, As, Cd, Cu, In, Mn, Mo, Tl	AES	Rasmy (1981)
>500	>1555				

[a] AAS — atomic absorption spectroscopy, AES — arc atomic emission spectroscopy, ICP-MS — inductively couple plasma mass spectrometry, LA-ICP-MS — laser ablation inductively coupled plasma mass spectrometry, LA-ICP-OES — laser ablation inductively coupled plasma optical emission spectroscopy, OES — optical emission spectroscopy, PIXE — proton-induced x-ray emission spectroscopy, TIMS — thermal ionisation mass spectrometry, WD-XRF — wave-length dispersive x-ray fluorescence spectroscopy.
[b] No information available on analytical technique(s) used.

selection of Ag, Cd, Co, Cu, Fe and Mn in addition to Ga, Ge and In. The concentration ranges reported for these elements in older studies using bulk analytical methods are virtually identical to those reported in recent studies using micro-analytical techniques (LA-ICP-MS, PIXE). Therefore, treating these different types of data together should not be problematic, as long as correction is made for the differences in sample volume. This was done by considering sample means rather than individual point analyses for data collected by micro-analytical techniques.

Except for Ag and Cu, for which there is some uncertainty, all of these nine elements are generally present as substitutions in the sphalerite lattice, and not as mineral inclusions (e.g. Cook et al., 2009; Belissont et al., 2014; George et al., 2015a, b). The presence of mineral inclusions

Table 2
Thermometric data on ore-forming fluids for selected deposits in the sphalerite database.

Deposit/Mine	District, country	T (°C)[a,b]	Salinity[a] (wt.% $NaCl_{eq}$)	Method	Mineral(s)	Type	Reference(s)
85 mine	Lordsburg district, New Mexico, USA	220 (180–260)	4.5 (3.0–6.0)	FIM	qtz, ca, fl	VEIN	Agezo (1995)
Abakaliki	Benue Valley, Nigeria	146 (126–166)	20 (15–25)	FIM	sph	MVT	Olade, Morton (1985)
Alston Moor	North Pennines, UK	110 (90–130)	22 (20–24)	FIM	fl, qtz, ank, dol, ca, sph, ba	MVT	Bouch et al. (2006)
Amelia mine	Upper Mississippi Valley, USA	123 (105–135)	20 (18–23)	FIM	sph	MVT	McLimans (1977); Rowan and Goldhaber (1996)
B Limestone	Bingham district, Utah, USA	347 (275–486)	–	FIM, SIP	FIM: qtz SIP: sph, gal, py	HTHR	Field and Moore (1971)
Binna Valley	Wallis, Switzerland	239 (212–272)	16 (6.4–25)	FIM	qtz	–	Klemm et al. (2004)
Bleiberg	Austria	120 (80–160)	21 (18–24)	FIM, PE	fl	MVT	Zeeh and Bechstädt (1994), Kappel and Schroll (1982)
Bok Su	Hwanggangri District, S. Korea	300 (240–360)	7.0 (3.5–11)	FIM	N/A	VEIN	So and Yun (1992)
Bushy Park	Griqualand West, South Africa	145 (110–180)	14 (11–17)	FIM	sph, dol	MVT	Kesler et al. (2007)
Cananea-Duluth mine	Arispe District, Sonora, Mexico	300 (250–350)	–	FIM	qtz	VEIN	Bushnell (1988), Meinert (1982)
Carlisle mine	Steeple Rock District, New Mexico, USA	260 (240–280)	2.4 (1.0–3.4)	FIM	qtz, sph, fl	VEIN	McLemore (1993)
Continental mine	Central District, New Mexico, USA	370 (320–420)	25 (10–40)	FIM	qtz	HTHR	Abramson (1981)
Dabaoshan	Qujiang District, Guangdong, China	300 (270–330)	–	FIM	qtz	HTHR	Huang et al. (1987)
Deardorff mine	Illinois–Kentucky Fluorspar district, USA	145 (130–160)	20 (19–21)	FIM	fl	MVT	Richardson and Pinckney (1984)
Empire Zinc mine	Fierro-Hanover District, New Mexico	337 (300–375)	10 (3–18)	FIM	N/A	HTHR	Turner and Bowman (1993)
Fankou	Guangdong, China	154 (135–173)	–	FIM	ca, qtz, sph	SHMS	Lu (1983)
Force Crag	Lake District, UK	120 (110–130)	24 (23–25)	FIM	fl, sph	VEIN	Stanley and Vaughan (1982)
Galena and Holden Fissures	Bingham District, Utah, USA	315 (243–430)	–	FIM, SIP	FIM: qtz SIP: sph, gal, py	VEIN	Field and Moore (1971)
Groundhog mine	Central District, New Mexico, USA	320 (290–340)	7.5 (6.2–8.5)	FIM	gt, px, qtz, sph	VEIN/ HTHR	Meinert (1987), Hawksworth and Meinert (1990)
Huize	Yunnan, China	187 (165–210)	8.6 (10.7–6.5)	FIM	ca	MVT	Han et al. (2004); Han et al. (2007)
Hutson mine	Illinois–Kentucky Fluorspar District, USA	125 (98–150)	20 (17–23)	FIM	fl	MVT	Spry and Fuhrmann (1994)
I Yeon Hwa	Taebaeksan District, South Korea	305 (270–340)	–	FIM	qtz, sph	HTHR	Yun and Einaudi (1982), Koh et al. (1992)
Jauken	Gailtaler Alpen, Austria	110 (90–130)	10 (7.9–12)	FIM	sph	MVT	Zeeh et al. (1998)
Jinding	Yunnan, China	180 (160–200)	9.5 (8.0–14)	FIM	sph, qtz	SHMS	Xue et al. (2007)
Kamioka	Hida Metamorphic Belt, Japan	330 (280–380)	7.0 (3.0–11)	FIM	sph	HTHR	Mariko et al. (1996)
Keel Prospect	Longford County, Ireland	157 (124–190)	11 (8.0–14)	FIM	sph, qtz	MVT	Wilkinson (2010)
Lark vein	Bingham District, Utah, USA	310 (294–330)	13 (6.0–20)	FIM	sph, qtz	VEIN	Roedder (1971)
Luziyuan	South China	280 (200–360)	–	FIM	N/A	HTHR	Xia et al. (2005)
Magmont mines	Viburnum Trend, Missouri, USA	107 (75–119)	21 (18–24)	FIM	sph	MVT	Appold and Wenz (2011)
Mineral Hill mine	Mineral Hill District, Nevada, USA	240 (220–260)	7.0 (3.0–10)	FIM	qtz, ba	HTHR	Vikre (1998)
Monte Cristo	Northern Arkansas, USA	115 (105–125)	23 (22–24)	FIM	qtz, sph	MVT	Leach et al. (1975), Stoffell et al. (2008)
Naica	Camargo District, Chihuahua, Mexico	305 (237–369)	37 (31–43)	FIM	fl	HTHR	Erwood et al. (1979), Sawkins (1964)
Nenthead	North Pennines, UK	108 (93–130)	22 (20–24)	FIM	sph, qtz	MVT	More et al. (1991)
Niujiaotang	South China	120 (101–142)	13 (9.2–16)	FIM	sph	MVT	Ye et al. (2012)
Oswaldo No. 2 mine	Central District, New Mexico, USA	350 (300–400)	20 (10–30)	FIM	qtz	HTHR	Ahmad and Rose (1980)
Pering	Griqualand West, South Africa	205 (200–210)	25 (10–30)	FIM	sph, qtz	MVT	Huizenga et al. (2006), Schaefer et al. (2004)
Picher Orefield	Tri-State District, Oklahoma/Kansas, USA	100 (80–120)	15 (10–20)	FIM	sph, dol, ca	MVT	McKnight and Fisher (1970), Stoffell et al. (2008)

Table 2 (*continued*)

Deposit/Mine	District, country	T (°C)[a,b]	Salinity[a] (wt.% $NaCl_{eq}$)	Method	Mineral(s)	Type	Reference(s)
Rosia Montana	S. Apuseni Mts., Romania	235 (200–270)	1.8 (0.2–3.4)	FIM	qtz	VEIN	Wallier et al. (2006), Manske et al. (2006)
Sacarimb	S. Apuseni Mts., Romania	245 (215–275)	3.5 (0.3–6.7)	FIM	qtz, sph	VEIN	Alderton and Fallick (2002)
Saint Salvy	Tarn, France	110 (80–140)	24 (23–25)	FIM	qtz	VEIN	Munoz et al. (1994)
Sam Bo	Shiheung District, S. Korea	210 (160–265)	9.0 (2.0–17)	FIM	qtz, sph, fl, ca	VEIN	So et al. (1984)
Shin Ye Mi	Taebaeksan District, S. Korea	325 (285–365)	8.0 (5.0–10)	FIM	sph	HTHR	Yang et al. (1993)
Snailbeach	Shropshire, UK	150 (130–170)	23 (21–25)	FIM	sph, fl, ca, ba, qtz	VEIN	Pattrick and Bowell (1991)
Thompson-Temperly	Upper Mississippi Valley, USA	123 (105–135)	20 (18–23)	FIM	sph	MVT	Rowan and Goldhaber (1996)
Threlkeld	Lake District, UK	120 (110–130)	24 (23–25)	FIM	fl, sph	VEIN	Stanley and Vaughan (1982)
Toyoha	Northern Japan	240 (200–275)	1.7 (0.8–2.6)	FIM	qtz, sph	VEIN	Yajima and Ohta (1979), Shikazono (1974)
Type B veins	Alcudia Valley, Spain	180 (150–210)	4.0 (3.5–4.5)	FIM	qtz	VEIN	Palero et al. (2003)
Type C veins	Alcudia Valley, Spain	220 (200–250)	7.2 (6.5–7.9)	FIM	qtz	VEIN	Palero et al. (2003)
Type D veins	Alcudia Valley, Spain	250 (200–300)	9.5 (6.5–12)	FIM	qtz, sph	VEIN	Palero et al. (2003)
Type E veins	Alcudia Valley, Spain	110 (101–121)	20 (10–30)	FIM	ank, ba	VEIN	Palero et al. (2003)
Various	Coeur d'Alene district, Idaho, USA	300 (250–350)	7.5 (5–10)	FIM	qtz	VEIN	Leach et al. (1988)

Abbreviations: FIM — fluid inclusion microthermometry; PE — petrographic evidence; SIP — sulphur isotope partitioning; ank — ankerite; ba — barite; ca — calcite; dol — dolomite; fl — fluorite; gal — galena; gt — garnet; py — pyrite; px — pyroxene; qtz — quartz; sph — sphalerite.

[a] Intervals given in parentheses correspond to either interval stated in source, or median ± one standard deviation of the data.

[b] Temperatures cited are generally homogenisation temperatures of fluid inclusions in various minerals and should therefore be considered *minimum* fluid temperatures.

might represent a particular problem when bulk mineral concentrates are analysed. Since most of the analytical data was collected in this way (cf. Section 2.1, Table 1), this is an aspect deserving further attention.

First, it is worth noting that the authors of the relevant studies generally took great care in the preparation of their mineral concentrates, including the estimation of impurity concentrations from polished grain mounts (e.g. Burnham, 1959; Rose, 1967; So, 1977). Reported purities are usually 99+ vol.% sphalerite. Second, even if inclusions are present a distinction should be made between different genetic types due to their different implications for data interpretation. Namely, inclusions formed by exsolution should be distinguished from those due to other mechanisms such as complex (e.g. dendritic) intergrowths or mineral–fluid reactions. Since we are interested in the original dissolved load of the sphalerite, inclusions formed by exsolution do not represent a particular problem. In fact, they should be included in the overall analysis, because otherwise the original composition of the sphalerite will not be captured correctly.

The high purity of the mineral concentrates analysed in the studies used in this article means that only very fine intergrowths between sphalerite and other minerals have not been separated. In the majority of cases, these will be due to exsolution or exsolution-like textures, such as chalcopyrite disease (Ramdohr, 1975). Exsolution-like textures of other minerals similar to chalcopyrite disease are also well known, involving e.g. the fahlores, stannite, pyrrhotite (Ramdohr, 1975) or even roquesite (Cook et al., 2011). Particularly the formation mechanism of chalcopyrite disease has long remained contentious. While older authors favoured an origin by exsolution (e.g. Ramdohr, 1975), newer studies generally propose a reaction of Fe-rich sphalerite with later Cu-rich fluids (Bortnikov et al., 1991) or some other partial replacement process (Barton and Bethke, 1987; Bente and Doering, 1993). It is not within the scope of this article to resolve this issue. We restrict ourselves to noting that these kinds of inclusions will mostly be a problem for the reported concentrations of Cu (and Ag if fahlores/galena are present), but should not have a major effect on measured Cd, Co, Ga, Ge, In and Mn concentrations, because these elements are generally not present in the included minerals (chalcopyrite, fahlores, pyrite) at sufficiently high concentrations (e.g. Burnham, 1959; George et al., 2015b). Concentrations of Fe in sphalerite are usually higher than 1 wt.%, and therefore the presence of inclusions at a level below 1 vol.% will generally not have a major effect on measured concentrations.

Despite the uncertainties with respect to their occurrence, we decided to include Cu and Ag in our analysis. This is because we believe that even if they are only partially present in solid solution their measured concentrations should still record interesting signals (cf. Cook et al., 2009). A detailed discussion of the consequences of the two potential modes of occurrence of Cu and Ag for data interpretation is provided with the detailed discussion of our results in Section 4.

2.5. Metamorphic overprinting

It is well known that the deformation of sulphide ores at moderate to high temperatures and pressures often results in extensive recrystallisation of the ductile ore minerals, including sphalerite (Clark and Kelly, 1973; Ramdohr, 1975; Cook et al., 1993). The consequent re-equilibration will inevitably cause changes to mineral chemistry (e.g. Kullerud, 1953; Hutchison and Scott, 1981). Because the probability that any particular ore deposit is affected by these kinds of processes depends largely on its tectonic setting, there is a different incidence rate for different deposit types: while most MVT, VEIN and HTHR deposits are unaffected, metamorphic overprinting and deformation are commonly observed in SHMS and VHMS deposits. This is illustrated by the metamorphic grade data presented for SHMS deposits in Table 4. At least half have been metamorphosed to lower greenschist facies or higher, meaning they have experienced metamorphic temperatures significantly above their range of formation (cf. Table 3). A similar compilation for VHMS deposits would yield comparable results but is not shown for reasons of brevity. Metamorphism has important

Table 3
Deposit type classification used for data analysis, as well as summary of general characteristics of ore-forming fluids and processes.

Deposit type	Typical host lithologies	Common minerals[†]	Formation temperature (°C)	Fluid Salinity (wt.% $NaCl_{eq}$)	Zinc concentration in fluids (ppm)	pH	Depth of formation (km)	Type of fluid	Metal source(s)	Precipitation mechanism
MVT Mississippi Valley-type	Carbonate-rich sediments[b]	Sph, gal, fl, qtz, dol	90–150[a,b]	10–30[b]	*Modern formation waters*[*]: 0.1–90[c–f] *Fluid inclusions*: 3–12[g]	4.5–6[h]	0.2–1.5[b]	Basinal brines,[b] Evaporitic brines[b]	Basement rocks, weathered regolith, basal sandstones, carbonate aquifers[b]	Fluid mixing[a,i]
SHMS Sediment-hosted massive sulphide	Clastic sediments[b]	Py, ba, sph, gal, qtz, cpy	90–250[j–m,¶]	10–30[j–m,¶]	N/A[¶]	N/A[¶]	0–0.5[b]	Basinal brines,[b] Evaporitic brines[b,m]	Basement rocks, weathered regolith, basal sandstones, carbonate aquifers[b]	Fluid mixing, cooling[b]
VHMS Volcanic-hosted massive sulphide	Mixed sedimentary–volcanic sequences[n]	Py, cpy, sph, gal, qtz, ba	*Modern*[#]: 150–400[n] *Ancient*: 150–350[o,p]	*Modern*[#]: 1–8[n] *Ancient*: 5–11[o,p]	*Modern*[#]: 1.7–5.2[n] *Ancient*: <20–430[o,p]	*Modern*[#]: 2.7–3.8[n]	0–0.5[n]	Seawater ± magmatic fluids[n]	Volcanic and sedimentary rocks, possible magmatic component[n]	Fluid mixing, cooling[n]
VEIN[§] Vein-type	Variable[q,r]	Variable[q,r]	*Coeur d'Alene*: 250–350[s] *Cornwall cross courses*: 120–145[t] *Morococha*: 250–350[u]	*Coeur d'Alene*: 5–10[s] *Cornwall cross courses*: 25–27[t] *Morococha*: 2–16[u]	*Coeur d'Alene*: N/A *Cornwall cross courses*: 50[t] *Morococha*: 100–1000[u]	*Coeur*: N/A *Cornwall*: <7[t] *Morococha*: <7[u]	*Coeur d'Alene*: 3.5–10[s] *Cornwall*: <2.0[t] *Morococha*: 0–2.0[u]	*Coeur d'Alene*: Metamorphic[s] *Cornwall*: Basinal brines[t] *Morococha*: Magmatic[u]	*Coeur d'Alene*: Basement[s] *Cornwall*: Sandstone aquifers[t] *Morococha*: Magma[u]	*Coeur d'Alene*: Cooling, P-drop[s] *Cornwall*: Rock reaction, fluid mixing[u] *Morococha*: Cooling, boiling[u]
HTHR High-temperature hydrothermal replacement	Carbonate-rich rocks[v]	Ca–Mg–silicates, Sph, gal, cpy, py, po, mgt[v]	250–400[v–x]	5–30[v–x]	*Bismark, Mexico*: 150–450[w] *El Mochito, Honduras*: 6000[x]	<7[v]	0.2–2.0[v–x]	Magmatic[v–x]	Magma[v–x]	Cooling, reaction with country rocks[v–x]

References: a — Sangster et al. (1994); b — Leach et al. (2005); c — Carpenter et al. (1974); d — Kharaka et al. (1987); e — Connolly et al. (1990); f — Stueber and Walter (1991); g — Stoffell et al. (2008); h — Emsbo (2000); i — Corbella et al. (2004); j — Gardner and Hutcheon (1985); k — Bresser (1992); l — Polito et al. (2006); m — Leach et al. (2004); n — Hannington et al. (2005); o — Zaw et al. (1996); p — Zaw et al. (2003); q — Cox and Singer (1986); r — Simmons et al. (2005); s — Leach et al. (1988); t — Stoffell et al. (2004); u — Catchpole et al. (2015); v — Meinert et al. (2005); w — Baker et al. (2004); x — Samson et al. (2008).

[†] Mineral abbreviations: ba — baryte, cpy — chalcopyrite, dol — dolomite, fl — fluorite, gal — galena, mgt — magnetite, po, pyrrhotite, py — pyrite, qtz — quartz, sph — sphalerite.

[*] Range of median values for data given by different sources.

[¶] Fluid characteristics not well constrained due to lack of data, and difficulty of interpretation due to post-formation history. Generally thought to be similar to MVT fluids (Leach et al., 2005).

[#] Confidence interval of geometric mean, estimated from collection of data in Hannington et al. (2005).

[§] VEIN deposits are a highly heterogeneous group, and it is therefore difficult to provide summary statistics for them. The three examples provided are intended to illustrate this variability.

Table 4
Metamorphic grade of SHMS deposits included in the chemical database.

Locality	Continent	Country	Metamorphic grade	Reference
Arzberg-Haufenreith	Europe	Austria	Greenschist facies	Weber (1990)
Silberbergstollen	Europe	Austria	Greenschist facies	Weber (1990)
Übelbach	Europe	Austria	Greenschist facies	Weber (1990)
Meggen	Europe	Germany	Zeolite facies or lower	Gasser (1974)
Rammelsberg	Europe	Germany	Prehnite–Pumpellyite facies	Large and Walcher (1999)
Bleikvassli	Europe	Norway	Upper amphibolite–lower granulite facies	Lockington et al. (2014)
Mofjellet	Europe	Norway	Amphibolite facies	Lockington et al. (2014)
Saxberget	Europe	Norway	Amphibolite facies	Vivallo and Rickard (1990)
Red Dog	North America	Alaska, USA	Zeolite facies	Leach et al. (2004)
Sullivan	North America	BC, Canada	Greenschist facies	Ethier et al. (1976)
Kimberley	North America	BC, Canada	Greenschist facies	Ethier et al. (1976)
Mount Isa	Australia	Australia	Greenschist facies	Lockington et al. (2014)
Broken Hill	Australia	Australia	Granulite facies	Lockington et al. (2014)
Fankou	Asia	China	N/A	Lu (1983)
Jinding	Asia	China	N/A	Ye et al. (2011)
Bainiuchang	Asia	China	N/A	Ye et al. (2011)
Gaobanhe	Asia	China	Not metamorphised	Li and Kusky (2007)
Tuogou	Asia	China	N/A	Xia et al. (2003)
Xitieshan	Asia	China	Greenschist facies	Wu (1985)
Yutang	Asia	China	N/A	Cheng et al. (2011)
Dabaoshan	Asia	China	N/A	Ye et al. (2011)
Dongchuan	Asia	China	N/A	Huichu et al. (1991)

implications for the interpretation of results for these two deposit types. This will be covered in more detail in Section 4.5 below.

2.6. Random sub-sampling (declustering)

Data points were grouped according to locality. Because the number of samples per locality is highly variable across the dataset (ranging from 1 to 92), individual localities were randomly sub-sampled to give a set including only one sample per locality. This procedure removes statistical bias towards the better sampled localities and ensures a homogeneous variance structure. The total number of observations (equal to the number of localities) for each element in each deposit type remaining after sub-sampling is shown in Table 5. Considering their global importance, both SHMS and VHMS deposits are clearly under-represented, while the opposite is true for VEIN deposits.

2.7. Data transformation and treatment of values below detection limit

Raw concentrations were log-transformed to ensure approximate normality required for statistical treatment (Fig. 1). This is a standard procedure in the treatment of trace element data (e.g. Gasser, 1974; Belissont et al., 2014) because the low concentrations can be considered to be measured on a ratio scale (Stevens, 1946). However, the often significant proportion of values below detection limit (BDL) is problematic (cf. Table 5). Because the logarithm of zero is minus infinity, the 'standard' treatment of replacing BDL values with zero is not possible for log-transformed data. Complete exclusion of such values from the analysis, on the other hand, would lead to both the loss of valuable information as well as a significant positive bias (cf. van den Boogaart and Tolosana-Delgado, 2013).

A number of different approaches exist to deal with this problem (van den Boogaart and Tolosana-Delgado, 2013). The simplest solution is imputation with a fixed value equal to the respective detection limit, or a fixed fraction thereof. Such treatment will generally result in an overestimation of the central tendency (mean), and an underestimation of the variance of a given sample population (van den Boogaart and Tolosana-Delgado, 2013). Another possibility is random imputation of the raw values with a normal distribution with both mean and standard deviation equal to the detection limit (van den Boogaart and Tolosana-Delgado, 2013). Negative values resulting from this imputation procedure may be adjusted simply by changing their sign. This approach has the advantage of giving a more realistic estimate of the variance while still resulting in an overestimate of the mean. The effects of both

Table 5
Number of observations for different elements in sphalerite from different deposit types.

	Proportion of global Zn resources (%)[a]	Ag	Cd	Co	Cu	Fe	Ga	Ge	In	Mn	Total no. of localities
MVT	17.4	91 (12)[b]	104 (1)[b]	72 (27)[b]	73 (2)[b]	81 (0)[b]	99 (1)[b]	105 (6)[b]	70 (43)[b]	90 (23)[b]	116
SHMS	23.4	11 (1)[b]	20 (0)[b]	9 (3)[b]	8 (0)[b]	15 (0)[b]	18 (0)[b]	10 (4)[b]	19 (3)[b]	12 (0)[b]	22
VHMS	25.6	27 (7)[b]	39 (0)[b]	16 (2)[b]	18 (0)[b]	15 (0)[b]	26 (2)[b]	24 (10)[b]	29 (3)[b]	17 (1)[b]	39
VEIN	6.7	168 (4)[b]	200 (1)[b]	150 (34)[b]	87 (0)[b]	81 (1)[b]	202 (59)[b]	194 (104)[b]	201 (73)[b]	184 (32)[b]	208
HTHR	13.5	65 (3)[b]	77 (0)[b]	63 (21)[b]	19 (0)[b]	26 (0)[b]	75 (32)[b]	68 (52)[b]	75 (12)[b]	68 (0)[b]	78
Total	86.6	362 (27)[b]	440 (2)[b]	310 (87)[b]	205 (2)[b]	218 (1)[b]	420 (94)[b]	401 (176)[b]	394 (134)[b]	371 (56)[b]	463

[a] Data from Penney et al. (2004).
[b] No. of values below detection limit (BDL).

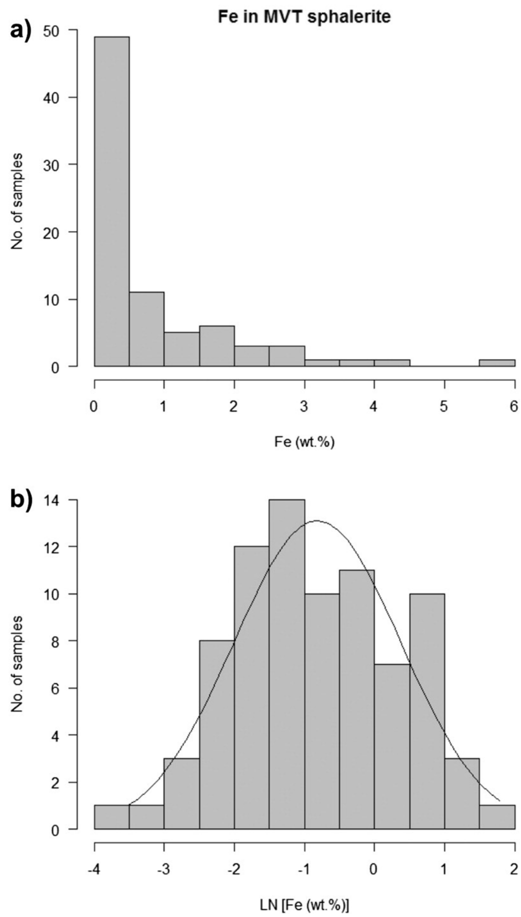


Fig. 1. Histograms of measured Fe concentrations in sphalerite samples from MVT deposits illustrating the effect of log-transformation: a) raw data, b) log-transformed data. The solid line in b) shows the shape of a normal distribution with the same mean and standard deviation as the log-transformed data. The highly skewed nature of the raw data is clearly apparent in a), while b) shows that log-transformed data closely approaches univariate normality.

fixed value and random imputation are illustrated in Fig. 2. It is obvious from this figure that random imputation (Fig. 2b) yields a more realistic approximation to the underlying distribution. Therefore, this method was used to treat all BDL values. It is worth noting at this point that because of the tendency of this procedure to overestimate the mean in cases where many BDL values are present, it should result in a *reduction* in the observed differences between different sample populations. This means that any significant differences which are still found in such cases are probably *more pronounced* in reality than indicated by the present analysis. The same is true for correlation trends.

2.8. Inspection of marginal distributions

For initial inspection of the data, estimates of the geometric means and corresponding 95% confidence intervals were computed for each element and every deposit type, omitting values missing at random (MAR, see below). While non-overlapping confidence intervals might already provide a clue to the existence of a significant difference between the means of two populations, their occurrence is not a sufficient criterion to make such a distinction (Browne, 1979; Payton et al., 2003). A one-way anova test (cf. Chambers et al., 1992) was therefore run for each element, with deposit type as the categorical variable and log-transformed element concentration as the measurement variable. For those elements for which this test returned a significance level of $p < 0.05$, the null hypothesis (that there is no difference) was rejected and Dunnett's modified Tukey–Kramer (DTK) test (Dunnett, 1980) was used to determine which pairs of deposit types are significantly different. Again, a significance level of $p < 0.05$ was used. The results of the DTK test allow us to group deposit types according to the concentrations of the various elements in sphalerite.

In addition, estimates were made of the proportion of the total variance of individual elements due to the variability *within* individual localities. This was done by averaging unbiased estimates of the standard deviation (Cureton, 1968) for those localities from which several samples were taken, and comparing the square of this average to the total variance. Knowing the partitioning of the variance is useful for understanding the origin of the observed overall variability. Is it due to processes operating within individual ore-forming systems, or to differences between these systems? This can provide clues about the controls on the observed differences and might be useful for the planning of future studies of global distribution trends.

2.9. Principal component analysis and treatment of values missing at random

When dealing with multidimensional analytical datasets, appropriate multivariate statistical techniques such as principal component analysis (PCA) should be used for data inspection because they are best suited to highlight the most relevant trends without external supervision (e.g. Koch, 2012). This works well for homogeneous datasets (e.g. Winderbaum et al., 2012; Belissont et al., 2014). However, in heterogeneous datasets values missing at random (MAR) may cause significant problems because incomplete observations cannot be accommodated by standard PCA procedures (e.g. Stanimirova et al., 2007). In the present case, MAR values are values missing because the element in question was not measured. This affects 25% of all entries in the final data matrix (1046 out of a total of 4167 entries, cf. Table 5). Several approaches exist to deal with this problem (cf. Stanimirova et al., 2007; Templ et al., 2011):

1) Removal of all incomplete observations: this is perhaps the least satisfying approach since it often results in a significant loss of information. For instance, even if only one out of nine values in an observation is missing, the whole observation would have to be removed. In the present case, such a procedure would result in a loss of 70% of the original data, which is not acceptable.
2) Column-wise imputation of missing values with the mean of the observed values: while this procedure does not necessitate the removal of any data-points, it has the disadvantage of not preserving the variance structure of the data (e.g. Prantner, 2011). Therefore, it is not suitable if a PCA is to be run on the imputed datasets.
3) Multi-variate imputation: a number of methods has been developed for multi-variate imputation. They follow three basic approaches — distance-based methods, covariance-based methods, and model-based methods. Model- and covariance-based methods are generally the best choice since they are designed to preserve the (variance) structure of the dataset. Potential disadvantages arise from the sensitivity of many of these methods to outliers, being based on the assumption of a multivariate normal distribution of the data. However, robust methods have been developed to overcome these problems (e.g. Templ et al., 2011).

Because of its clear advantages compared to the other methods, multi-variate model-based imputation was chosen for the present analysis. In order to avoid the potential effects of outliers, the IRMI (iterative model-based imputation using robust methods) algorithm developed by Templ et al. (2011) was used. Initial guesses of the missing values were taken to be equal to the median of the corresponding data-column, and random noise was added to the final best estimate of each value to preserve randomness. Data for each deposit type was treated separately. Imputations were found to converge well for all datasets.

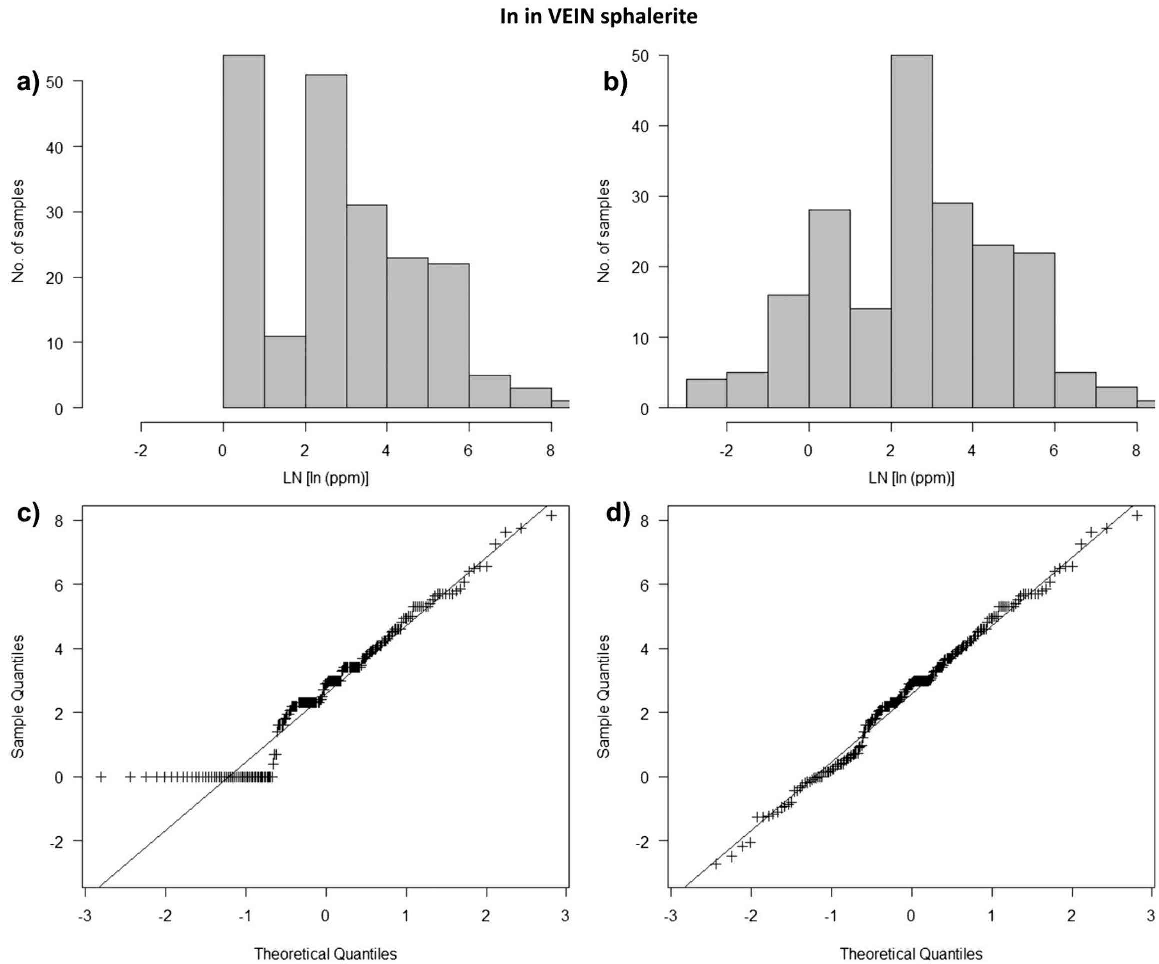


Fig. 2. Histograms and corresponding normal Q–Q plots of log-transformed In concentrations in VEIN sphalerite illustrating the effects of fixed-value and random imputation of BDL values on the sample population: a) histogram of In concentrations using fixed-value imputation, b) histogram of In concentrations using random imputation as described in Section 3.3, c) normal Q–Q plot corresponding to a), and d) normal Q–Q plot corresponding to b). It is clearly seen that fixed-value imputation results in a highly skewed and asymmetric distribution, while random imputation results in a nice bell curve closely approaching normality (as shown by the Q–Q plot in d).

Datasets for individual deposit types were then compiled into one big dataset, and individual data columns normalised to have mean 0 and standard deviation 1. A PCA was then run on this large dataset and the results used to assess the higher-dimensional nature of the compositional differences in sphalerite samples from different deposit types.

Last but not least, the relationships between the different PCs and temperature and salinity of the ore-forming fluids were tested for the subset of deposits where such information was available. A more detailed description of these tests is given in Section 3.4 below because it relies on the results of the PCA. The degree to which the concentrations of individual elements correlate with temperature and salinity of the ore-forming fluids was also tested.

The R software environment (R Development Core Team, 2004) with additional packages DTK (Lau, 2013) and VIMGUI (Schopfhauser et al., 2014) was used for data analysis.

3. Results

Before presenting the results of the statistical analyses, it is instructive to convey an impression of the significant degree of scatter present in the data, and the resultant overlap between different sample populations. This is best done using a series of boxplots (Fig. 3). While it should be apparent from these plots that large overlaps between different deposit types exist for every element, it is also clear that statistically significant differences are present between some of them (e.g. between HTHR and MVT deposits). The geometric means and confidence intervals shown in Table 6 corroborate this first impression. Table 7 gives the estimated log-standard deviation for each element in each deposit type, while Table 8 provides estimated proportions of the total variance due to the differences between samples from the same locality. Two things should be noted: 1) for each element, the log-standard deviation for each deposit-type is, with few exceptions (e.g. Cd in MVT deposits), very similar to the other deposit-types and 2) the proportion of the total variance due to intra-locality variability is generally small (≤30% of the total variance). The two notable exceptions to the latter observation are Cu (particularly in VHMS deposits) and Ge in HTHR deposits. The high value for Cu in VHMS sphalerite is probably an artefact of the very limited number of observations, while for Ge in HTHR sphalerite it is due to the large proportion of values below detection limit. This results in an underestimation of the between-deposit variance (cf. low value for log-standard deviation in Table 7). The generally small size of intra-locality variance compared to overall variance, even within type categories, means that the features described in the following paragraphs mostly reflect differences between individual deposits, rather than differences between samples.

3.1. One-way anova and DTK tests

The results of the one-way anova tests (Table 9) clearly indicate that statistically significant differences exist for all elements except Cd. Even if the conservative Bonferroni correction for the evaluation of multiple

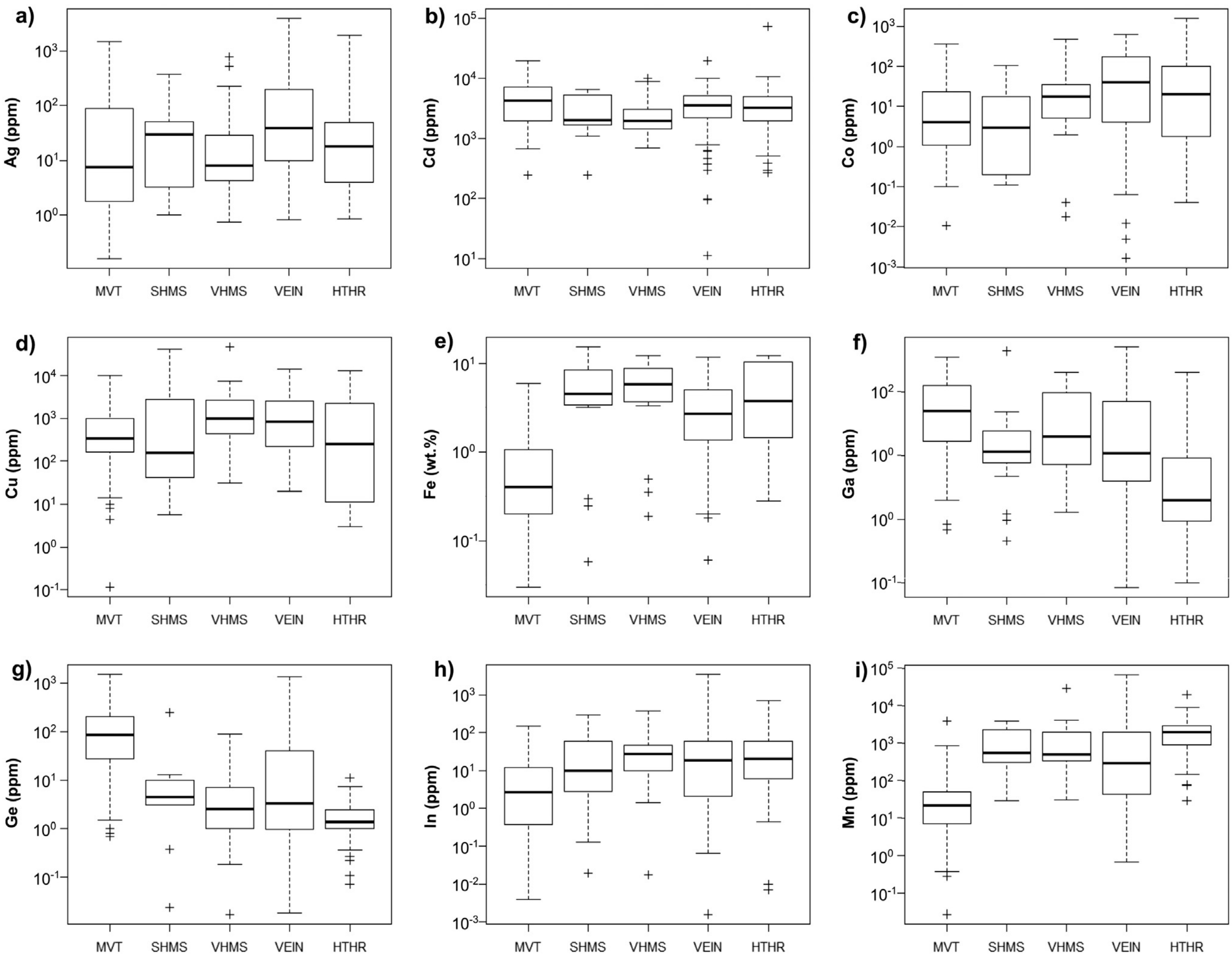


Fig. 3. Boxplots of log-transformed concentrations in different deposit types for the nine elements considered in the statistical analysis: a) silver, b) cadmium, c) cobalt, d) copper, e) iron, f) gallium, g) germanium, h) indium, and i) manganese. Note large overlaps between the sample populations corresponding to different deposit types.

comparisons (i.e. dividing the required p-value by the number of tests conducted, in the present case yielding a maximum value of $p = 0.006$, cf. Dunn, 1961) is applied, the p-values for all other elements are still significant. However, the results of the DTK tests indicate that, in addition to Cd, no significantly different pair of deposit types exists for Cu. This discrepancy in the results is due to the fact that the one-way anova test assumes homoscedasticity of the data (i.e. equal variance within different populations) while the DTK test does not.

A graphical summary of the results of the DTK tests for Ag, Fe, Ga, Ge, In and Mn is shown in Fig. 4. Each panel corresponds to one element and shows the geometric means for the different deposit types, either in decreasing or increasing order away from MVT. The results for Co are not included in this figure, since only one significant pairwise difference was found (between MVT and VEIN deposits).

The general pattern of element behaviour is strikingly regular: the means of five (Fe, Ga, Ge, In, Mn) out of the six elements order in the sequence MVT–VEIN–HTHR. Additionally, even though the large uncertainties associated to the means of element concentrations in SHMS and VHMS sphalerite preclude an unambiguous positioning of these two deposit types on the general trend, the sequence: MVT–SHMS–VHMS is observed in four (Fe, Ge, In, Mn) out of the five cases where the sequence MVT–VEIN–HTHR is observed. Silver is the only element

Table 6
Geometric means of element concentrations in sphalerite (first row) and corresponding 95% confidence intervals (second row; *italics*).

	Ag (ppm)	Cd (ppm)	Co (ppm)	Cu (ppm)	Fe (wt.%)	Ga (ppm)	Ge (ppm)	In (ppm)	Mn (ppm)
MVT	12	3600	6	350	0.44	42	63	2.2	18
	7–19	*2700–4800*	*3–11*	*210–560*	*0.34–0.58*	*32–56*	*45–89*	*0.9–5.3*	*11–28*
SHMS	16	2400	2	310	3.0	11	3.7	10	590
	5–52	*1700–3400*	*0.2–23*	*38–2500*	*1.3–7.0*	*5–22*	*0.5–26*	*3–33*	*250–1400*
VHMS	13	2100	10	1,100	3.7	19	2.2	22	730
	6–28	*1800–2600*	*2.3–40*	*530–2400*	*1.9–7.3*	*11–35*	*0.9–6*	*10–50*	*340–1600*
VEIN	42	3200	21	690	2.3	14	5	14	250
	31–57	*2800–3600*	*13–33*	*490–1000*	*1.8–2.9*	*10–19*	*3–8*	*9–20*	*170–370*
HTHR	15	2900	14	140	3.4	3.1	1.4	16	1,600
	10–24	*2400–3600*	*6–32*	*37–530*	*2.1–5.4*	*1.8–5.3*	*0.9–2.2*	*9–27*	*1200–2100*

Table 7
Estimated log-standard deviations (for natural logarithms).

	Ag (ppm)	Cd (ppm)	Co (ppm)	Cu (ppm)	Fe (wt.%)	Ga (ppm)	Ge (ppm)	In (ppm)	Mn (ppm)
MVT	2.3	1.5	2.1	2.0	1.2	1.4	1.7	2.3	1.8
SHMS	1.9	0.8	2.8	3.0	1.6	1.5	2.4	2.4	1.5
VHMS	1.7	0.6	2.7	1.6	1.3	1.4	1.7	2.1	1.5
VEIN	1.9	0.9	2.6	1.7	1.1	1.8	2.3	2.2	2.3
HTHR	1.7	0.9	2.6	2.9	1.2	1.8	0.9	2.1	1.2

whose behaviour deviates from the general trend: namely, sphalerite from VEIN deposits has a higher mean Ag concentration than sphalerite from both MVT and HTHR deposits (Fig. 4a). Overall, the regularity of the trends for Fe, Ga, Ge, In and Mn appears to suggest a one-dimensional pattern of the differences in the trace element content of sphalerite between deposit types, defined by a decrease in Ga and Ge concentrations and a concomitant increase in Fe, In and Mn concentrations in the two short sequences MVT–VEIN–HTHR and MVT–SHMS–VHMS.

3.2. Covariance structure of the dataset

As a first step in the identification of relevant relationships in a multivariate dataset it is often helpful to examine its covariance structure. In order to do this we compiled the (imputed) data from different deposit types into a single dataset and calculated the corresponding correlation matrix (Table 10). Highly significant positive correlations (correlation coefficient $R > 0.30$, $p < 10^{-12}$) were found for the pairs Fe–Mn, Ga–Ge, Cu–In, Cu–Ag and Fe–In, while similarly significant negative correlations exist for Ge–Mn, Ge–Fe, Ga–Mn and Ga–Fe. These relationships are broadly consistent with the trends in mean concentrations observed for the different deposit types, i.e. the decrease in Ga and Ge concentrations accompanied by an increase in Mn, Fe, and to a lesser degree In concentrations, going from MVT to HTHR deposits. Other relationships (Cu–In, Cu–Ag) apparently have little relevance for the differences between deposit types.

Using a suitably modified version of the correlation matrix as a dissimilarity matrix, the similarity in the behaviour of the different trace elements can be visualised in a cluster dendrogram (van den Boogaart and Tolosana-Delgado, 2013). This may be used to identify groups of elements showing similar behaviour. Different possibilities exist for the conversion of a correlation matrix to a dissimilarity matrix. The results for two different transformations are shown in Fig. 5, each with its respective merits.

In Fig. 5a, dissimilarity was taken as $(1 - R)$. This results in the grouping together of those elements which are positively correlated, while those which are negatively correlated are grouped apart. It is clear from this figure that three groups of positively correlated elements may be distinguished: Ga and Ge show strong positive correlation, as do Fe and Mn (cf. Table 10), while Ag, Cd, Co, Cu and In fall into a group with relatively weak positive correlations among the different members.

In Fig. 5b, dissimilarity was taken as $(1 - |R|)$. This results in the grouping together of those elements connected by the strongest relationships, irrespective of their positive or negative sign. Again, three groups of elements result: the first contains Ga, Ge, Fe and Mn all of which show strong pair-wise correlation. The second contains Ag, Cu and In, for which correlations are somewhat weaker. Finally, the third group contains Cd and Co which neither correlate strongly with each other, nor with any of the other elements.

3.3. Principal component analysis (PCA)

Fig. 6 summarises the results of the PCA. Although only 47% of the total variance is captured by principal components (PCs) 1 and 2, biplots including any of the other seven PCs are not presented, since they do not convey any additional information. From the biplots and histograms shown in Fig. 6a and b it is apparent that pronounced differences between deposit types only occur along PC 1, but not PC 2. Furthermore, none of the other seven PCs shows differences between deposit types. The HTHR and MVT populations appear to form the two end-members of the overall trend and show minimal overlap (Fig. 6b), while VEIN deposits span the whole range of possible values, with a mean leaning slightly closer towards the HTHR end of the spectrum. VHMS and SHMS deposits seem to be significantly more similar to HTHR than MVT deposits, but in contrast to HTHR deposits show a somewhat larger degree of overlap with the MVT field.

The loadings of PC 1 indicate that the elements making the largest contributions to the described differences between deposit types are Fe, Ga, Ge and Mn, all of which appear to be of approximately equal importance. Indium on the other hand appears to be only about half as important as any one of these four elements, and the importance of Ag, Cd, Co and Cu is negligible. These results are in perfect accordance with those from the analysis of the marginal distributions presented in Section 4.1. They also confirm the earlier suggestion that the differences between deposit types follow a simple one-dimensional pattern. That is, the trace element content of sphalerite from different deposit types only differs substantially along one of the 9 dimensions of the sample space considered here, and this dimension is described by PC 1.

Considering the circle of correlations depicted in Fig. 6c the three broad groups of elements already identified in Fig. 5a are reproduced. They show the following behaviour: The concentration of elements in group I (Fe and Mn) decreases towards the MVT end of the spectrum, while the concentration of elements in group III (Ga and Ge) increases in the same direction. Finally, the concentration of elements in group II (Ag, Cd, Co, Cu and In) either remains unaffected by changes in deposit type (Cu, Cd), or tends to decrease only slightly towards the MVT end of the spectrum (Ag, Co, In).

Table 8
Estimated proportions of total variance due to intra-locality variability (%).

	Ag	Cd	Co	Cu	Fe	Ga	Ge	In	Mn
MVT	12	6	8	13	22	32	21	22	10
SHMS	8	8	33	6	4	13	3	9	20
VHMS	10	14	8	~100	3	30	3	9	11
VEIN	31	12	11	38	34	15	15	21	15
HTHR	29	11	11	41	17	23	63	30	28

Table 9
Results of one-way anova tests.

Element	p-Value
Ag	3.9×10^{-6}
Cd	0.051
Co	0.002
Cu	0.003
Fe	$<2 \times 10^{-16}$
Ga	$<2 \times 10^{-16}$
Ge	$<2 \times 10^{-16}$
In	2.1×10^{-8}
Mn	$<2 \times 10^{-16}$

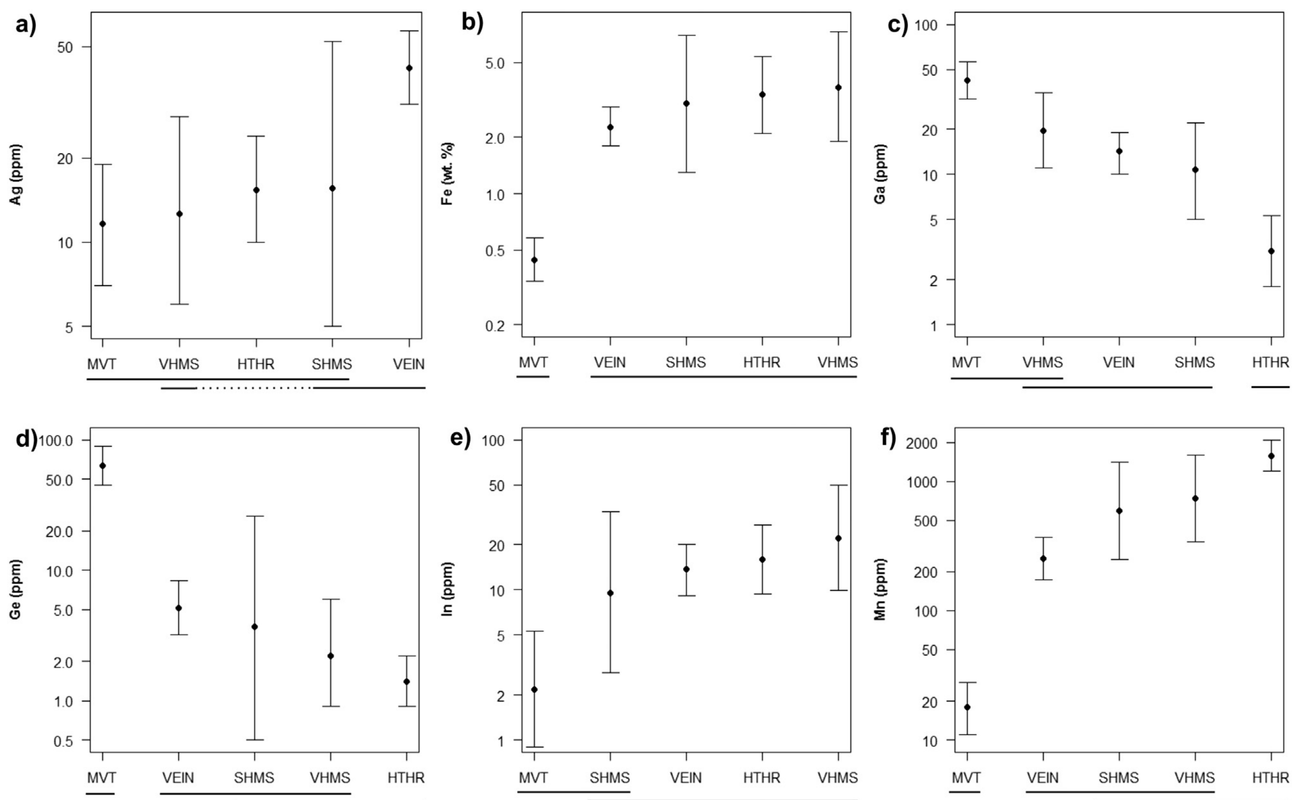


Fig. 4. Graphical representations of the results of DTK tests for a) Ag, b) Fe, c) Ga, d) Ge, e) In and f) Mn. In each panel, deposit types are listed from left to right according to the geometric mean of the respective element's concentration in sphalerite, with concentrations either decreasing or increasing away from MVT. Plots show geometric means and 95% confidence intervals on the y-axis (logarithmic scale). The lines at the bottom of each panel connect deposit types the means of which are not significantly different according to the DTK test statistic. The dotted line in panel a) indicates that while VHMS, SHMS and VEIN deposits are in one group, HTHR deposits do not belong to this group.

Since only Fe, Ga, Ge, Mn, and to a lesser degree In, are important contributors to PC 1, a non-normalised approximation to this coordinate might be computed as:

$$\text{PC 1}^* = \ln\left(\frac{C_{\text{Ga}}^{0.22}\cdot C_{\text{Ge}}^{0.22}}{C_{\text{Fe}}^{0.37}\cdot C_{\text{Mn}}^{0.20}\cdot C_{\text{In}}^{0.11}}\right) \tag{1}$$

where the concentrations of Ga, Ge, Mn and In are in units of ppm, and the concentration of Fe is in units of wt.%. The coefficients are equal to the loading of each element divided by its log-standard deviation. Fig. 7a shows that PC 1* is as effective in discriminating sphalerite from MVT and HTHR deposits as PC 1. Sphalerite from VEIN, SHMS and VHMS deposits also plots in similar relative positions to the two end members as before (Fig. 7b, cf. Fig. 6a). It should be noted, however, that particularly the position of SHMS deposits in the general sequence of MVT–VEIN–SHMS–VHMS–HTHR suggested by these plots is relatively uncertain, due to the small number of observations from this deposit type (cf. Table 5).

Last but not least, it is interesting to note with reference to Fig. 5 that the elements making the largest contributions to the first three PCs clearly reflect the groupings seen in the cluster dendrogram with (1 − |R|) as the dissimilarity measure (Fig. 5b). The main contributors to PC 1 are Fe, Ga, Ge and Mn, while for PC 2 they are Ag, Cu and In, and for PC 3 they are Cd and Co (cf. Fig. 6). This is not a coincidence and serves to further illustrate the utility of cluster dendrograms in showing the covariance structure of a multivariate dataset.

Table 10
Correlation matrix for the complete (imputed) chemical dataset of 463 deposits.

	Ag	Cd	Co	Cu	Fe	Ga	Ge	In
Cd	0.04							
Co	*0.14*	*−0.13*						
Cu	**0.36**	*0.22*	*0.14*					
Fe	*0.13*	−0.05	*0.23*	0.06				
Ga	0.05	*0.11*	*−0.11*	*0.22*	**−0.32**			
Ge	0.00	0.01	−0.06	0.07	**−0.39**	**0.51**		
In	*0.17*	0.02	*0.11*	**0.42**	**0.33**	0.04	*−0.17*	
Mn	0.03	0.07	−0.08	0.03	**0.52**	**−0.38**	**−0.56**	*0.21*

Numeric values correspond to R. Values in italics correspond to relationships with p-values between 0.05 and 1×10^{-12}, while all values in bold have $p < 1\times10^{-12}$. The latter correspond to the most significant relationships present within the dataset.

3.4. Relationships between principal components and fluid temperature and salinity

By definition, each of the PCs of a multivariate dataset is orthogonal to all others (Koch, 2012). Therefore, each PC may be thought of as reflecting the action of an independent external control, or set of controls. Plots of the different PCs against relevant explanatory variables representing these controls may therefore shed light on the underlying mechanisms governing the structure of the overall dataset. In the present case, only the temperature and salinity of the ore-forming fluids could be tested as potential explanatory variables, and only for a relatively small subset of the chemical data (~10%, cf. Section 2.2).

In order to do these tests, we first devised approximations for the different PCs, similar to that shown for PC 1 in Eq. (1) above. As a rule, only those elements whose coefficients (loading over log-standard deviation, cf. Fig. 6) are larger than ~1/6th of the highest coefficient were included. The results for PC 2* and PC 3* are:

$$\text{PC 2}^* = \ln\left(C_{\text{Cu}}^{0.34}\cdot C_{\text{Ag}}^{0.22}\cdot C_{\text{In}}^{0.19}\cdot C_{\text{Cd}}^{0.19}\cdot C_{\text{Ga}}^{0.18}\cdot C_{\text{Ge}}^{0.07}\cdot C_{\text{Co}}^{0.07}\cdot C_{\text{Fe}}^{0.07}\right). \tag{2}$$

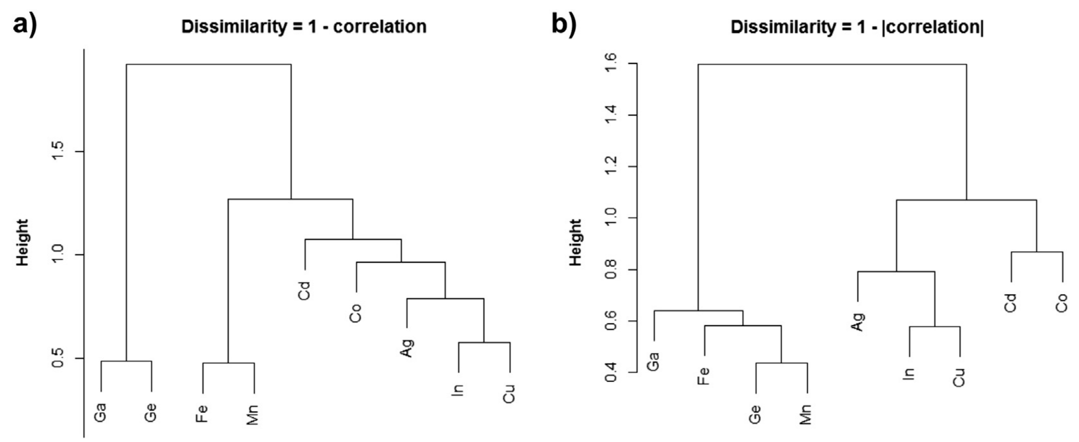


Fig. 5. Cluster dendrograms showing the association of variables for the whole imputed sphalerite dataset (463 deposits). These were obtained using the Ward algorithm (cf. van den Boogaart and Tolosana-Delgado, 2013) on two different kinds of dissimilarity matrix derived from the correlation matrix of the dataset: a) using $d = 1 - correlation$ as the dissimilarity measure, b) using $d = 1 - |correlation|$ as the dissimilarity measure. The different interpretations of the resultant dendrograms are discussed in the main text.

$$\text{PC } 3^* = \ln\left(\frac{C_{\text{Cd}}^{0.60} \cdot C_{\text{Mn}}^{0.12}}{C_{\text{Co}}^{0.27} \cdot C_{\text{Fe}}^{0.09}}\right). \quad (3)$$

The reason for using these approximations is twofold: first, they allow us to focus on the most relevant elements for each PC. Second, they are more practical to use since not all elements are required for the accurate estimation of each PC.

It was mentioned earlier (Section 2.2) that the completeness of the set of analysed elements was taken into account in the selection of deposits for the temperature and salinity dataset. This was done using the approximations devised above: only those deposits for which at most one element was missing for each PC* were included.

The means and standard errors of the different PCs* were then estimated for each deposit. The standard errors take into account both sampling and imputation errors. By sampling errors we mean those errors arising from the variability of sphalerite compositions within single samples and across a deposit. Plots were than made of the PCs* against temperature and log-transformed salinity, and simple linear and quadratic fits were used to test for the strength of each relationship. The results for the first three PCs* are shown in Fig. 8, along with the best fits (i.e. those with the lowest p-value). Plots involving the other six PCs* are not shown because they would not have added any relevant information.

Two highly significant relationships ($p < 0.01$) could be identified. Namely, PC 1* correlates strongly with the minimum temperature of the ore-forming fluids (Fig. 8a), while PC 2* correlates with salinity (Fig. 8d). Although the latter relationship was fitted with a quadratic equation to allow for a simple assessment of R^2 and p, it might be better represented by a combination of two linear trends intersecting at a log-salinity of approximately 2.1.

At the $p < 0.05$ level, the relationship between PC 1* and salinity is also significant (Fig. 8b). However, this appears to reflect the relationships between temperature and salinity (Fig. 9) and temperature and PC 1* (Fig. 8a), rather than an independent relationship between salinity and PC 1*. The trend towards lower salinity at higher temperatures seen for MVT and VEIN deposits on Fig. 9 is well known and represents the transition from classic MVT deposits to epithermal veins (Wilkinson, 2001). The reversal in this trend as temperatures increase further reflects the transition from epithermal to magmatic-hydrothermal deposits, i.e. skarns (Wilkinson, 2001).

3.5. Element concentrations in sphalerite as a function of fluid temperature and salinity

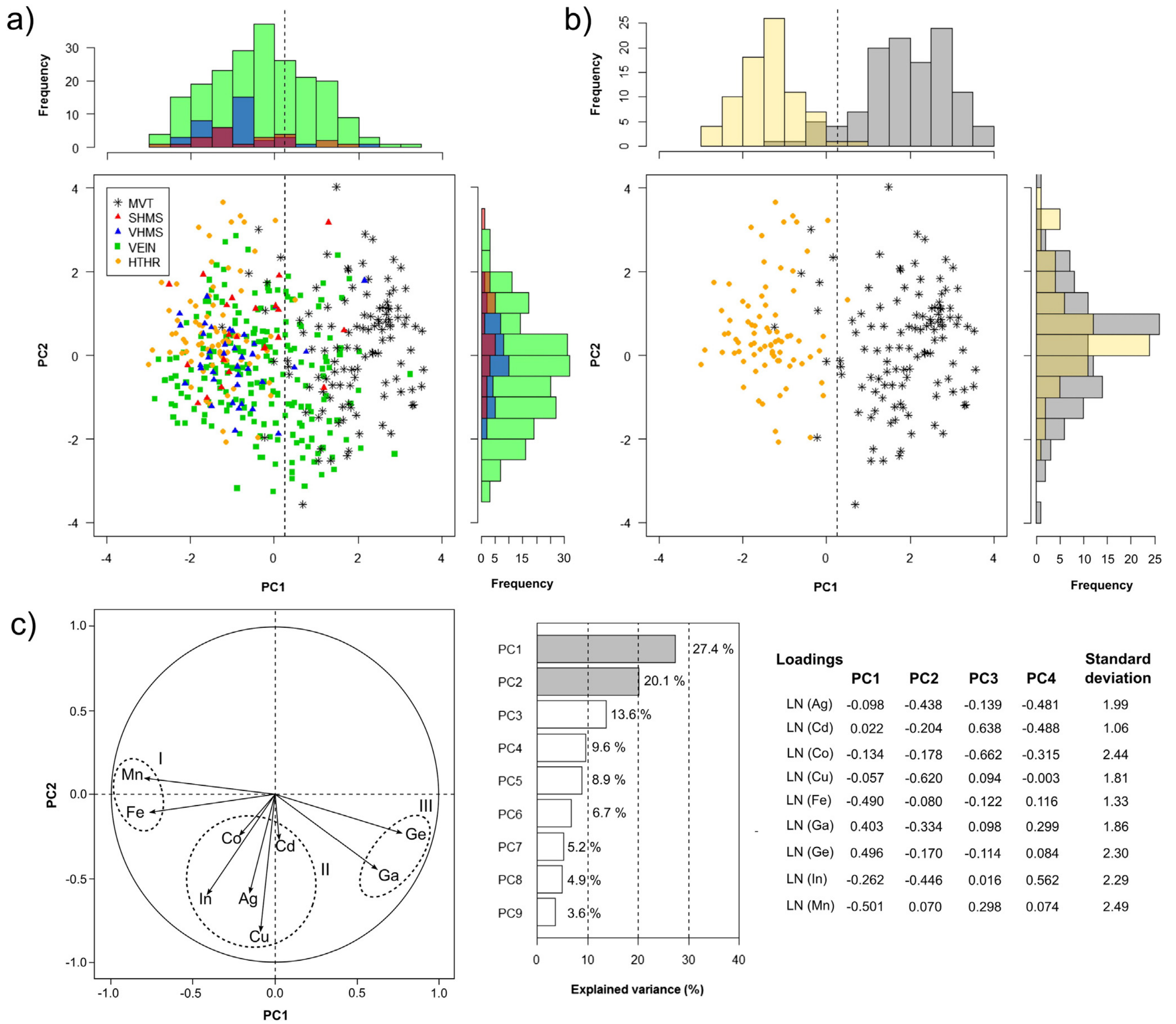
The proportions of the total variance captured by individual PCs are shown in Fig. 7. Unfortunately, this information does not allow for an assessment of how much of the total variance of each *element* is controlled by the different geological factors behind each PC. This is of particular interest for the explanatory variables identified in the previous subsection, i.e. fluid temperature and salinity. The strength of these relationships determines to what degree the concentration of any given element in sphalerite may be predicted from a knowledge of the temperature and salinity of the ore-forming fluids.

To address this point, the means of the log-transformed concentrations of each element were calculated for each of the 51 deposits listed in Table 2. Plots against fluid temperature and salinity were then fitted to linear and quadratic relationships, respectively. Additionally, linear fits were made for plots against PC 1* and PC 2* for all elements, both for the fluid dataset (51 deposits), as well as the complete dataset (463 deposits). A summary of all these fits is given in Table 11. The numbers correspond to R^2 values, with those highlighted in bold indicating significant relationships at the $p < 0.05$ level. Corresponding scatterplots of Ga, Ge and In concentrations against fluid temperature and PC 1* are shown in Fig. 10 by way of example. Similar plots for the other six elements are shown in Fig. B1 in Electronic Annex B.

First, we note that there is good correspondence between the R^2 values for the relationships of different elements with temperature on the one hand, and PC 1* on the other. Iron is an exception, but this might be due to the random effects which may occur for small datasets. This close correspondence in the strengths of the respective relationships is as expected if PC 1* is accepted as representing fluid temperature. It follows from the results in Table 11 that the strength of the relationship between concentrations and fluid temperature for individual elements decreases roughly in the following order: Mn ~ Ge > Ga ~ Fe > In > Ag ~ Cd ~ Co ~ Cu. For the last four elements, the relationship is not statistically significant.

Second, a similarly good correspondence in R^2 values as observed between temperature and PC 1* is not seen for salinity and PC 2*. It is, however, worth noting that the group of elements for which a significant relationship exists with salinity (Ag, Cu, Ga, Ge, In, Mn) is mostly identical to the group for which a significant relationship exists with PC 2* in the fluid dataset (Ag, Cu, Ga, In). The additional correlation of Ge and Mn with salinity might be attributable to the relationship between salinity and temperature noted before (Fig. 9). It is worth emphasising that most of the total variance seen in Cu concentrations is captured by PC 2* (>70%), while less is captured for In, Ag and Ga, in decreasing order. Although relationships between PC 2* and Cd, Co, Fe and Mn are also statistically significant for the complete dataset, they only account for a negligible fraction of the total variance in the concentrations of these elements and might therefore be ignored.

It will be noted that of the elements considered, only Cd and Co correlate with neither temperature nor salinity (nor, for that matter, PC 1*



	PC1	PC2	PC3	PC4	Standard deviation
LN (Ag)	-0.098	-0.438	-0.139	-0.481	1.99
LN (Cd)	0.022	-0.204	0.638	-0.488	1.06
LN (Co)	-0.134	-0.178	-0.662	-0.315	2.44
LN (Cu)	-0.057	-0.620	0.094	-0.003	1.81
LN (Fe)	-0.490	-0.080	-0.122	0.116	1.33
LN (Ga)	0.403	-0.334	0.098	0.299	1.86
LN (Ge)	0.496	-0.170	-0.114	0.084	2.30
LN (In)	-0.262	-0.446	0.016	0.562	2.29
LN (Mn)	-0.501	0.070	0.298	0.074	2.49

Fig. 6. Graphical representation of the results of the principal component analysis (PCA), showing: a) a covariance biplot of all data points, and corresponding histograms of PC 1 and PC 2 for the data points from SHMS, VHMS and VEIN deposits, b) a covariance biplot of only the MVT and HTHR data points with corresponding histograms to illustrate the marked difference between these two populations, c) the circle of correlations corresponding to the biplots in a) and b), and a legend including a scree-plot and details of the loadings of the different principal components (PCs). Note that the histograms in a) and b) are plotted as overlapping and not as stacked histograms, such that they accurately reflect the shapes of the respective distributions. MVT and HTHR data points were not included in the histograms in a) to avoid unnecessary cluttering. It should be obvious from panel b) that the MVT and HTHR populations are so different that they hardly show any overlap. However, this difference is restricted to PC 1. The means for the distributions of PC 2 values of both deposit types are virtually identical. Groups of elements showing similar behaviour are marked in c) as I, II and III. Note correspondence of this grouping with that seen in Fig. 5a. The angle between two arrows in c) is related to the covariance between the concentrations of the corresponding elements: if the angle is close to 0°, element concentrations correlate positively, if it is close to 90°, they do not correlate, and if it is close to 180° they correlate negatively.

or PC 2*). Instead, a significant part of their total variance (~40% for Cd and ~75% for Co) is captured by PC 3* for which a suitable explanatory variable is still needed. This is discussed in more detail below (Section 4.3).

4. Discussion

The results presented in the previous section show clearly that certain aspects of the composition of hydrothermal sphalerite correlate well with fluid temperature and salinity. The correlation with temperature also explains the differences observed between different deposit-types. In the following subsections we will first examine how the limitations of our dataset might affect the observed strength of these relationships. Additionally, we consider possible mechanisms by which these relationships might arise. We then discuss other factors expected to enact a control on sphalerite composition and how they might be reflected in our data. Based on the strong relationship between temperature and PC 1*, we suggest a tentative new sphalerite geothermometer and calibrate it using the fluid temperature data in Table 2. In the last two subsections we first use a number of examples to examine how metamorphic overprinting might have affected the compositions of SHMS and VHMS sphalerites, and finally discuss the general applicability of our results to the study of (hydrothermal) ore deposits.

4.1. Temperature

Bearing in mind the limitations of our dataset, namely that fluid temperature and salinity data were collected on different sample sets than

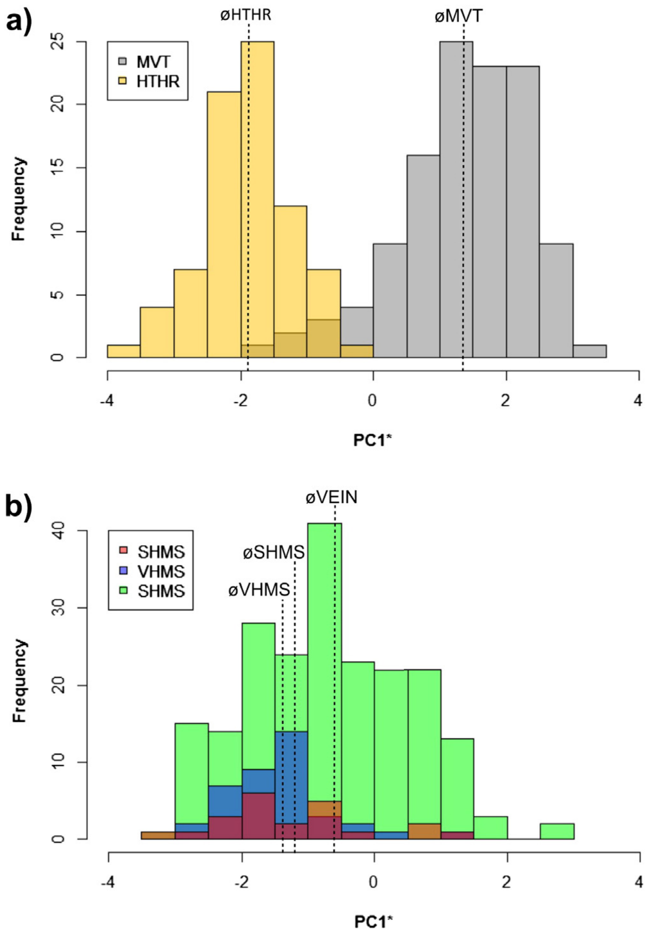


Fig. 7. Illustration of the suitability of PC 1* as an approximation to PC 1, showing: a) histograms of MVT and HTHR data points and b) histograms of SHMS, VHMS and VEIN data points. Note that histograms are again plotted as overlapping and not stacked histograms. Vertical lines labelled with the abbreviations of the different deposit types indicate the mean values of PC 1* for the respective sample populations. Note the great similarity between the SHMS and VHMS populations.

the chemical data, that sphalerite often does not contain any measurable fluid inclusions (although co-genetic minerals might), and finally, that temperatures might vary significantly over the extent of a single ore deposit, it is certainly striking to see how strong the relationship found between temperature and PC 1* is. About 80% of the total variance in PC 1* is explained by temperature alone.

To get a better idea of the true strength of the relationship, we estimated how much of the unexplained 20% of the total variance may be explained by the measurement and sampling errors arising from the nature of the data. This was done using Monte Carlo simulations as follows: a large number of perfectly correlated datasets were simulated. Individual measurements of temperature and PC 1* in each dataset were then perturbed by random normal errors with standard deviation equal to the respective mean standard error for the measurement (as estimated from the actual dataset). The median value for R^2 of the perturbed datasets in these simulations was found to be 0.90. That is, about half of the 20% of unexplained variance can be accounted for by measurement and sampling errors alone.

Since there might be other sources of error, such as systematic differences in concentration measurements between different studies, which could not be taken into account in our simulations because there is no means of quantifying them, we are confident that there is actually a 1:1 correspondence between PC 1 and temperature. That is, the true strength of the underlying relationship is very probably greater than that seen in Fig. 8a. Therefore, differences in the mean temperature of formation can explain virtually all the differences in sphalerite composition seen between the different deposit-types, because these differences are captured entirely by PC 1* (cf. Section 3.3). Furthermore, temperature can account for ~65% of the total variance in Mn and Ge concentrations, ~40% of the total variance in Ga concentrations, ~20–50% of the total variance in Fe concentrations and ~10% of the total variance in In concentrations (cf. Table 11).

An important question is what the mechanism behind these relationships is: Correlation does not necessarily imply causality (e.g. Wright, 1921). A third factor could control both temperature and PC 1* leading to the observed trend. For instance, increasing distance from a heat-source would lead to a decrease in temperature accompanied by changes in fluid composition (via e.g. equilibration with surrounding rocks, fluid mixing) which should be reflected in the composition of the sphalerite precipitated at the site of ore-formation. Other scenarios can also be envisaged.

However, the strength as well as the simplicity of the relationship between temperature and PC 1 might suggest that the main control behind PC 1 is thermodynamic, acting through changes in the chemical equilibrium constants for the relevant substitution and buffering reactions. The general mechanism is easily illustrated by considering a simplified reaction for the partitioning of a generic divalent trace element A between sphalerite and fluid:

$$\mathrm{ZnS}_{ss} + \mathrm{A}^{2+}_{liq} \leftrightarrow \mathrm{AS}_{ss} + \mathrm{Zn}^{2+}_{liq}. \tag{4}$$

The equilibrium constant for this reaction is (cf. McIntire, 1963):

$$K_A = \frac{[\mathrm{AS}_{ss}] \cdot \left[\mathrm{Zn}^{2+}_{liq}\right]}{[\mathrm{ZnS}_{ss}] \cdot \left[\mathrm{A}^{2+}_{liq}\right]} \approx \frac{(c_A/c_{Zn})_{ss} \cdot \left[\mathrm{Zn}^{2+}_{liq}\right]}{\left[\mathrm{A}^{2+}_{liq}\right]} \tag{5}$$

where the c's denote concentrations, and the subscript ss refers to the sphalerite solid solution. The approximation is valid for small concentrations of element A in sphalerite. The log-transformed concentration of element A in sphalerite, $\ln(c_{A,ss})$, is given by:

$$\ln(c_{A,ss}) = \ln(c_A/c_{Zn})_{ss} + \ln(c_{Zn,ss}). \tag{6}$$

If we assume that the ratio between the activities of Zn^{2+}_{liq} and A^{2+}_{liq} in solution is buffered externally by a material of constant composition (e.g. country-rock or rock present in fluid conduits), then:

$$\frac{\left[\mathrm{Zn}^{2+}_{liq}\right]}{\left[\mathrm{A}^{2+}_{liq}\right]} = K_{r,A} \cdot (c_{Zn}/c_A)_r \tag{7}$$

where (c_{Zn}/c_A) is the concentration ratio of Zn to element A in the rock buffer, and $K_{r,A}$ is the equilibrium constant describing the buffering reaction. Yardley (2005) demonstrated that the composition of natural hydrothermal solutions is mostly controlled by rock-buffering. Therefore, this represents a sensible assumption. Substituting Eqs. (5) and (7) into Eq. (6) we have:

$$\ln(c_{A,ss}) = \ln(K_A) + \ln(K_r) + \ln(c_{Zn}/c_A)_r + \ln(c_{Zn,ss}). \tag{8}$$

Because $\ln(c_{Zn}/c_A)$ and $\ln(c_{Zn,ss})$ are constant, the temperature dependence of $\ln(c_{A,ss})$ will be described entirely by the temperature dependence of $\ln(K_A)$ and $\ln(K_{r,A})$. The simplest form for the temperature dependence of an equilibrium constant, K_{eq}, of a chemical reaction can be derived from the van't Hoff equation as (Mortimer and Müller, 2001):

$$K_{eq} = \frac{A_{eq}}{T} + B_{eq} \tag{9}$$

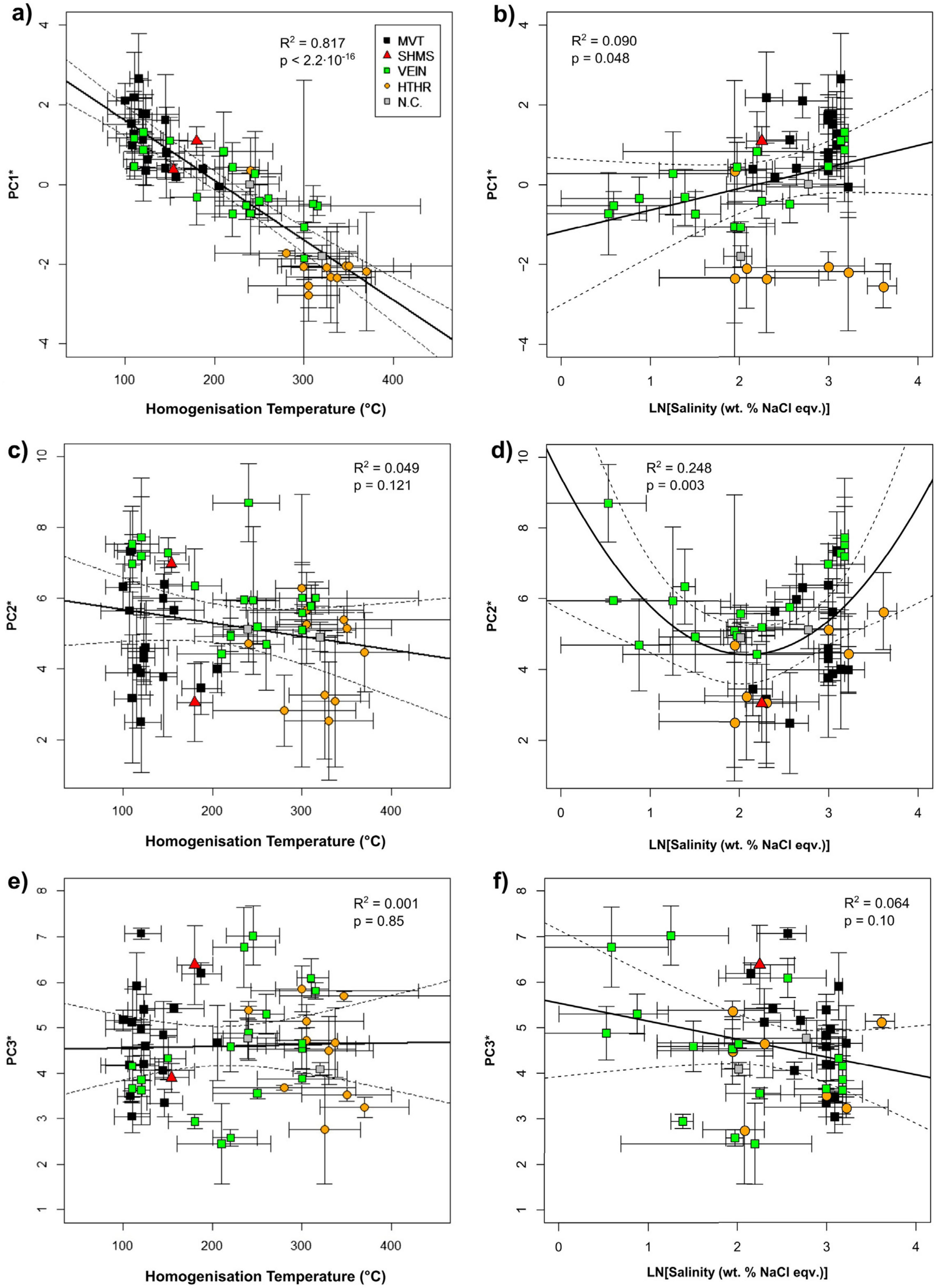


Fig. 8. Plots of approximations to the first three principal components (definitions in Eqs. (1), (2) and (3)) against homogenisation temperature and salinity of fluid inclusions: a) PC 1* vs. temperature, b) PC 1* vs. fluid salinity, c) PC 2* vs. temperature, d) PC 2* vs. fluid salinity, e) PC 3* vs. temperature, f) PC 3* vs. fluid salinity. Error bars denote 95% confidence intervals. The complete plot data for this figure may be found in digital form in Electronic Annex A. For temperature and salinity data also refer to Table 2. N.C. — not classified.

where both A_{eq} and B_{eq} are constants. This relationship is valid if both $\Delta H°$ and $\Delta S°$ for the reaction do not vary with temperature. While this assumption usually provides a good approximation over a small temperature range (e.g. McIntire, 1963), a more general form can be derived if it is instead assumed that $\Delta c°$, the difference in heat capacities between the products and reactants, is constant. We then have (e.g. Galaon and David, 2011):

$$K_{eq} = \frac{D_{eq}}{T^2} + \frac{E_{eq}}{T} + F_{eq}. \tag{10}$$

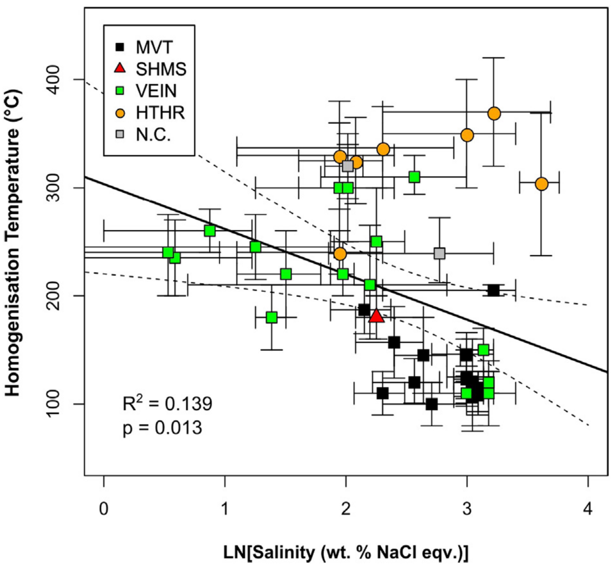


Fig. 9. Plot of fluid temperature against fluid salinity for the deposits shown in Fig. 8. Note similarity to Fig. 8b. Data from Table 2.

For $\ln(c_{A,ss})$, we would therefore generally expect a relationship with temperature of the following form:

$$\ln(c_{A,ss}) = \frac{(D_A + D_{r,A})}{T^2} + \frac{(E_A + E_{r,A})}{T} + (F_A + F_{r,A}) + \ln(c_{Zn}/c_A)_r + \ln(c_{Zn,ss}). \quad (11)$$

Depending on the relative magnitudes of the D_i and E_i, $\ln(c_{A,ss})$ might either decrease, increase or stay approximately constant as temperature is increased. Because PC 1 is a linear combination of log-transformed concentrations, a similar dependence on temperature would be expected if temperature-induced changes in chemical equilibrium are the main control mechanism. Fits for both the linear (Eq. (9)) and quadratic (Eq. (10)) form are shown in Fig. B2 in the Supplementary material (Electronic Annex B). A quadratic fit is found to provide a slightly better representation than a linear one. However, both appear to be adequate having similar values of R^2 (~0.80), and are not substantially different from the linear fit against temperature shown in Fig. 8a. Unfortunately, data on the temperature dependence of the counterparts to K_A and $K_{r,A}$ for most of the relevant trace elements is not available, such that further tests (by prediction) of the suggested equilibrium control on sphalerite compositions cannot be conducted.

Table 11
Correlation of trace element concentrations with temperature, salinity, PC1* and PC2*.

	Temperature	PC1* (FIM data)	PC1* (all data)	LN (salinity)	PC2* (FIM data)	PC2* (all data)
LN(Ag)	0.00	0.02	0.00	**0.16**	**0.37**	**0.35**
LN(Cd	0.03	0.01	0.00	0.05	0.00	**0.08**
LN(Co)	0.01	0.06	**0.01**	0.04	0.07	**0.06**
LN(Cu)	0.02	0.01	0.00	**0.15**	**0.77**	**0.70**
LN(Fe)	**0.20**	**0.46**	**0.57**	0.00	0.01	**0.02**
LN(Ga)	**0.37**	**0.42**	**0.43**	**0.16**	**0.39**	**0.19**
LN(Ge)	**0.65**	**0.66**	**0.64**	**0.14**	0.04	**0.04**
LN(In)	0.06	**0.11**	**0.14**	**0.15**	**0.61**	**0.37**
LN(Mn)	**0.67**	**0.59**	**0.66**	**0.14**	0.02	0.00

Note: numeric entries correspond to R^2 (= proportion of total variance explained by relationship) for the different row–column pairs; values in bold correspond to statistically significant relationships ($p < 0.05$).
FIM data — analytical data for deposits for which temperature data was also available (Table 3).
All data — data for all deposits in the database which were also included in the PCA (incl. imputed values).

We further note that a relationship similar to Eq. (11) would also result if we were to consider buffering by a fluid in which most Zn and A is not present as the simple aqueous species, Zn^{2+}_{liq} and A^{2+}_{liq}, but is instead bound in various complexes, the dissociation of which controls their activity ratio. Cooling of initially identical fluids to different degrees would result in changes in the dissociation constants of these complexes and therefore a dependence of the activity ratios on temperature, in a similar way as before. Fluids might initially be identical because they might have equilibrated with rocks of identical composition under similar p/T conditions. Such a scenario would assume no re-equilibration with the surrounding rock-mass during fluid transport and ore deposition. In reality, most ore-forming fluids probably follow a mixed scenario: much of their dissolved load originates from equilibration with a deep(er) source, but some re-equilibration with surrounding rocks also occurs during fluid transport and ore deposition. Such mixed scenarios would still be expected to yield the kind of relationship seen in Eq. (11).

Last but not least we note that our model clearly has its limitations: source compositions, particularly trace element content, might differ between deposits and this might have a stronger influence than temperature for some elements, providing one potential explanation for that part of the total variance not represented by PC 1. It will also be noted that for elements with a valence state other than 2+, relations become more complex and are crucially dependent on the type of substitution mechanism (cf. McIntire, 1963), i.e. whether charge balance is achieved by coupled substitution (e.g. Cu and In, Johan, 1988; Cook et al., 2012) or the creation of cation or anion vacancies in the sphalerite lattice (e.g. Ge, Cook et al., 2015, Belissont et al., 2016). This may introduce additional dependencies on the activity of Zn^{2+}_{liq}, the activities of other cations (in the case of coupled substitution), and sulphur fugacity, fS_2. However, a detailed discussion of these different mechanisms would go beyond the scope of this article. Therefore, we restrict ourselves to noting that most of these other factors will also be controlled by temperature, e.g. via the temperature dependence of the solubility product of sphalerite (e.g. Yardley, 2005).

We conclude that there is no good reason to *reject* thermodynamic control as an explanation for the relationship between PC 1 and temperature. It provides the simplest account for our observations and should therefore be favoured until evidence for greater complexity is forthcoming (Sober, 1994). This explanation might imply negligible influence of fluid source composition on PC 1.

4.2. Salinity

The observed relationship between salinity and PC 2 is certainly not as strong as the one between temperature and PC 1. It is also more complex, probably resulting from a combination of two contrasting linear trends (cf. Section 3.4). Naturally, the same limitations regarding measurement and sampling errors apply as for PC 1 above. However, they alone are not sufficient to account for the significantly weaker nature of the observed relationship and the significant proportion of unexplained variance.

Since PC 2 correlates most strongly with measured Cu concentrations (Table 11), the incorporation of mineral inclusions in the form of chalcopyrite disease represents an additional concern (cf. Section 2.4). If such inclusions occur in a random manner and do not reflect material formerly present in solid solution, their presence might be expected to introduce a large amount of background noise, obscuring any systematic signal present in dissolved concentrations. The same considerations apply to Ag concentrations. There is no reason why concentrations due to randomly occurring inclusions should record a signal in fluid salinity. It is clear, therefore, that the underlying relationship between salinity and a number similar to PC 2 could actually be much stronger than that observed. Unlike the case of temperature, however, it remains entirely unclear what the exact mechanism controlling this relationship might be. A complete explanation will likely involve a number of

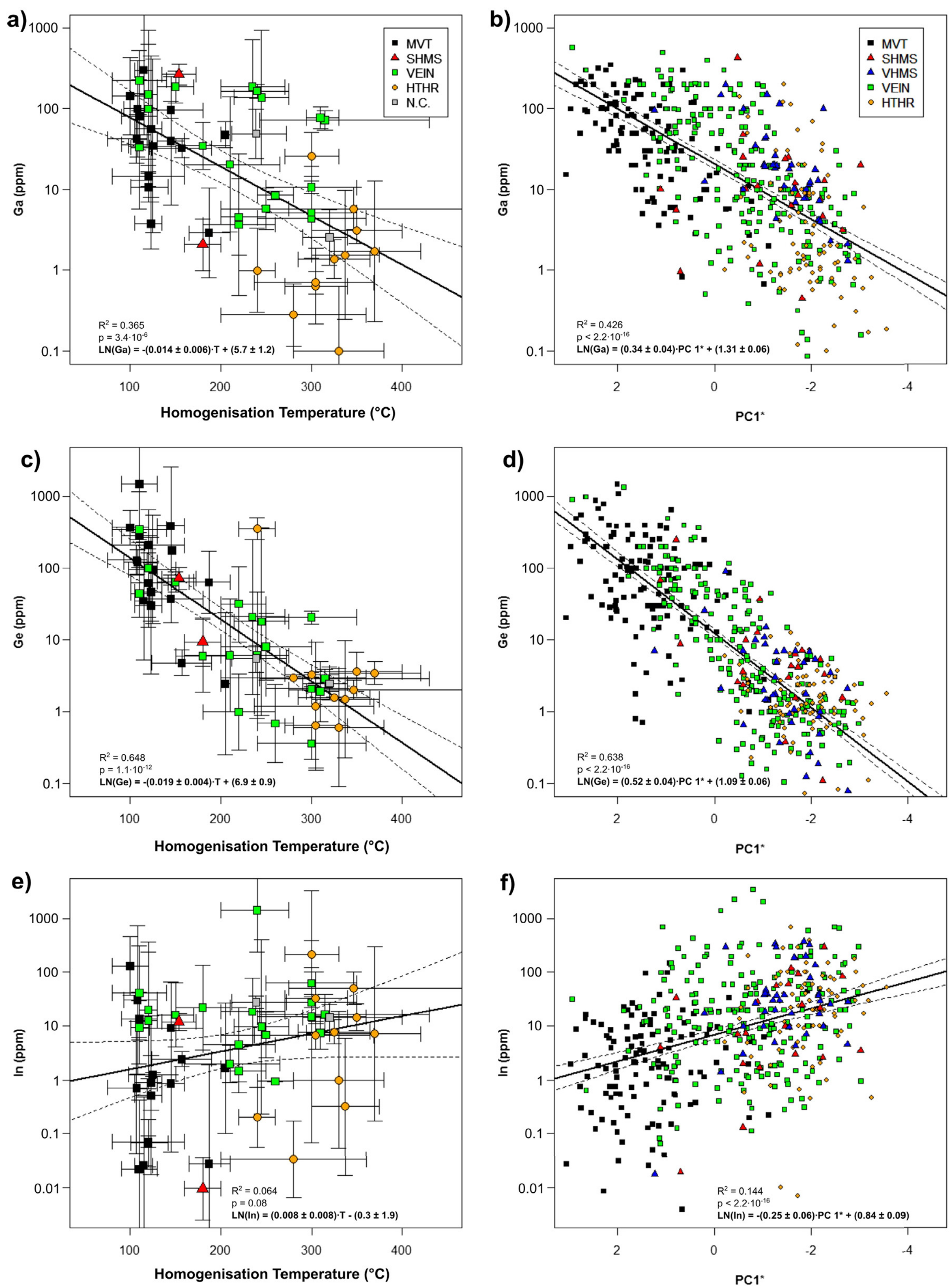


Fig. 10. Scatter plots showing the correlation of Ga, Ge, and In concentrations in sphalerite with temperature and PC 1*: a) Ga vs. temperature, b) Ga vs. PC 1*, c) Ge vs. temperature, d) Ge vs. PC 1*, e) In vs. temperature, f) In vs. PC 1*. Note the clear correspondence between the plots against temperature and those against PC 1*. Equivalent scatter plots for the other six elements are shown in Fig. B1 in Electronic Annex B.

different aspects, including: 1) the incorporation mechanism(s) for the affected elements, 2) the effect of the ionic strength of the fluid on the activity coefficients of individual elements (e.g. Debye and Hückel, 1923), 3) the effect of chloride activity on metal speciation (complex formation) (e.g. Wood and Samson, 2006), and 4) other factors which could independently control both fluid salinity and metal activities, such as differences in fluid sources. In particular, any explanation should be able to account for the non-linearity of the observed relationship which might indicate the action of different control mechanisms for low and high salinities.

For the reasons stated above, the major conclusions from this subsection must be that: 1) a relationship between fluid salinity and the trace element content of sphalerite probably exists and is likely stronger than the one observed in Figs. 8d, 2) a large part of the unexplained variance seen in PC 2 might reflect the effect of chalcopyrite and potentially other mineral inclusions, but could also simply reflect the Cu concentration in sphalerite, and 3) an explanation for the non-linear relationship between salinity and sphalerite chemistry is currently not possible.

4.3. Other controls on sphalerite chemistry

Referring back to Fig. 6, it should be clear that PC 1 and PC 2 account for 27 and 20% of the total variance, respectively. While the close correspondence between PC 1 and temperature means that about 25% of the total variance is explained by changes in the formation temperature of the sphalerite, the much weaker correlation between salinity and PC 2 means that only as much as 5–10% of the total observed variance may be explained by variations in fluid salinity. In total, this leaves 65–70% of the total variance unexplained. This remainder must be due to the action of other factors, as well as potential measurement errors and other sources of statistical noise. Concentrations of Cd and Co, and to a lesser degree Ag, Cu and In are particularly affected (cf. Table 11).

In the following, we will focus on the two most critical aspects of the unexplained variance, since a more detailed discussion is not warranted by the evidence at hand. Namely, we will consider first, the proportion of the variance in PC 2 not accounted for by variations in fluid salinity, and second, the potential significance of PC 3.

It was noted above that the good correlation between PC 2 and Cu suggests that this principal component might, in addition to the incorporation of Cu into the lattice, reflect the extent to which the relevant sphalerite is affected by chalcopyrite disease. If chalcopyrite disease reflects exsolution (Ramdohr, 1975), then its occurrence should correlate with the original Cu content of the sphalerite. If it reflects reaction with a later Cu-rich fluid (Bortnikov et al., 1991), on the other hand, its occurrence might simply record the likelihood of the occurrence of such a reaction in a given ore-forming system. In both cases, we might expect a correlation of the measured Cu concentration with Cu activity in the overall system and, therefore, the source of the ore-forming fluids. It is worth noting in this context that the majority of the observed Cu concentrations is below the expected solubility limit of Cu in sphalerite (~1000–10000 ppm, cf. Kojima and Sugaki, 1985).

Both the original incorporation of Cu as well as later reactions should also affect the concentrations of other trace elements in sphalerite. This effect might be responsible for some of the observed correlation of PC 2 with Ga and In concentrations (cf. Table 11). In this context it is worth emphasising the statistically significant correlation between Cu and In concentrations in the overall dataset (Table 10, Fig. 5) which might suggest the involvement of coupled substitution — a mechanism for the incorporation of In into sphalerite previously suggested by several authors (Johan, 1988; Cook et al., 2012). The same mechanism might, to a lesser degree, apply to the incorporation of Ga and explain its involvement in PC 2 and weaker, but still statistically significant, positive correlation with Cu concentration (cf. Table 10). The presence of these correlations actually suggests some degree of crystal chemical control and might therefore imply that most of the measured Cu concentrations are not simply due to the random incorporation of mineral inclusions. Whether chalcopyrite disease results from exsolution or reaction, Cu concentrations in affected sphalerite will still be elevated compared to unaffected sphalerite. This would be expected to aid the incorporation of Ga and In into the lattice. In stoichiometric terms, much more Cu is usually present than would be needed to balance the incorporation of trivalent Ga and In.

In contrast to PC 2, PC 3 mostly reflects changes in the concentrations of elements which can reasonably be assumed to be present in solid solution: Cd and Co (Cook et al., 2009; Belissont et al., 2014). Since there is no correlation between PC 3 and temperature or salinity, changes in the concentrations of these two elements must reflect the action of some other control mechanism, e.g. the composition of the source rock or aquifers (cf. Section 4.1). In addition, pH, Eh or fS_2 of the fluids can also not be ruled out as potential controls.

In conclusion, we propose that the composition of the ore-forming fluids, and in particular Cu activity as potentially captured by PC 2, might enact a strong influence on the In concentration, and a less strong influence on the Ga concentration of hydrothermal sphalerite. An interesting question for future research concerns the significance of variations in Cd and Co concentrations as captured by PC 3.

4.4. A new sphalerite geothermometer?

The strong correlation between PC 1 and homogenisation temperature suggests that this relationship might be usable as a geothermometer. A simple linear calibration of temperature against PC 1 is shown in Fig. 11a. While it was noted in Section 4.1 above that a better representation might be provided by a linear (or quadratic) fit of reciprocal temperature to PC 1, reference to Fig. B2 in the Supplementary material will show that this makes little difference over the temperature range of interest, i.e. about 100 to 400 °C. However, even if the calibrations in Fig. B2 are used we would caution against extrapolation beyond the range shown, as long as the thermodynamics of (or other controls on) the relationship between PC 1 and temperature are not well understood.

Application of the thermometer is illustrated by three examples from Table 2 in Fig. 11b. This shows that prediction of the mean (homogenisation) temperature is possible to within 50–70 °C.

It should be noted that this is merely a proposal. Only testing it in a large number of cases in which temperature is independently determined by other means will show whether it is useful as a tool in the study of ore deposits. If its applicability on the small as well as the large scale could be demonstrated, its key advantage over the use of fluid inclusions would be the capability to study processes occurring at much smaller time-scales. Should this geothermometer become widely accepted, we propose that it be called GGIMFis for **G**a, **G**e, **I**n, **M**n and **F**e **i**n **s**phalerite.

4.5. Effect of metamorphism on trace element contents in sphalerite

Before coming to the final evaluation of the general applicability of our results, it is necessary to devote some attention to the effects of metamorphic overprinting on the trace element content of sphalerite. It is well known that sphalerite will start to recrystallise and re-equilibrate with other sulphide minerals, even when heated only to moderate temperatures (Clark and Kelly, 1973; Ramdohr, 1975; Cook et al., 1993). Particularly the effects of this re-equilibration on Cu, Fe and Mn concentrations are well documented (Kullerud, 1953; Scott and Kissin, 1973; Hutchison and Scott, 1981; Kojima and Sugaki, 1984, Kojima and Sugaki, 1985; Lusk et al., 1993). The concentrations of all three elements are generally expected to increase with increasing temperature. Effects on other minor and trace elements are also expected, but are not well documented.

The potential effects of metamorphism were already noted in Section 2.5, but this has been deliberately ignored up to this point. It is particularly relevant in understanding the results for VHMS and SHMS deposits because these are generally affected by deformation-induced recrystallisation under variable metamorphic conditions. This is due to the tectonic settings in which they are formed and preserved.

Because metamorphism is essentially the re-equilibration of existing mineral assemblages under differing temperatures and pressures, we decided to restrict ourselves to the more detailed evaluation of its effects on PC 1. PC 1 is the most relevant because it describes the differences in sphalerite composition between different deposit types and correlates with formation temperature.

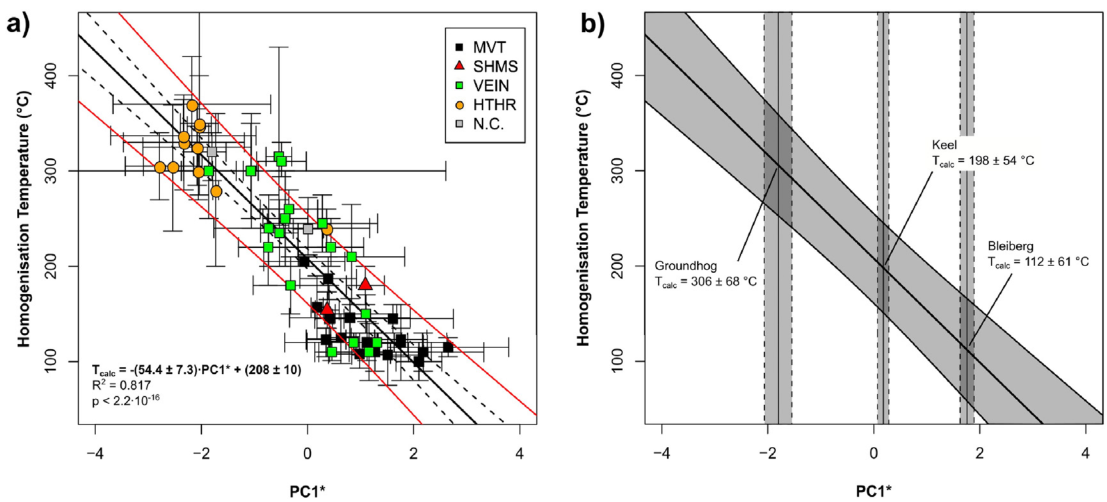


Fig. 11. The GGIMFis geothermometer: a) calibration using same data as shown in Fig. 8a, b) application to three example cases, Bleiberg, Austria, the Keel Prospect, Ireland, and the Groundhog mine, Central District, New Mexico. Compare predicted values for homogenisation temperatures to those listed in Table 2. Note that fits to plots of reciprocal temperature against PC 1* are shown in Fig. B2 in Electronic Annex B.

For this comparison, we compiled metamorphic grade data for those deposits in our database for which reasonably complete analytical datasets were available (Table 12). We then estimated the mean of PC 1* for each deposit, including confidence intervals, and converted these values to temperatures using the GGIMFis geothermometer (cf. Section 4.4). A plot of GGIMFis temperature against peak metamorphic temperature for the deposits in Table 12 shows that the data deviates strongly from a 1:1 relationship (Fig. 12). Namely, GGIMFis temperature increases with peak metamorphic temperature up to about ~310 °C (PC 1* ~ −1.8), and then remains constant as peak temperature is further increased. We interpret this to reflect closure of the sphalerite system at around ~310 °C during retrograde metamorphism. Diffusion (and therefore recrystallisation) rates in sulphides are generally much higher than they are in silicate minerals (Craig and Vaughan, 1994) and this should provide a reasonable explanation for such low closure temperatures. We further note that the closure temperature for sulphur isotope fractionation between sulphides, particularly sphalerite and galena, during retrograde metamorphism appears to be similar (e.g. Whelan et al., 1984; Lusk and Krouse, 1997), lending further support to this interpretation. It is worth particular mention that closure temperature is expected to be higher for faster cooling rates. Regional metamorphism usually involves relatively slow cooling, while ore-forming systems represent relatively short-lived thermal anomalies (Yardley and Cleverley, 2013) and therefore show higher cooling rates. The faster cooling rates in ore-forming systems mean that GGIMFis will also be applicable at higher temperatures in these cases.

The consequences of resetting by metamorphism for the observed values of PC 1 in affected deposits should be abundantly clear from Fig. 12. Namely, the mean value of PC 1* is reset to -1.8 ± 0.7 for all deposits affected by greenschist facies or higher grade metamorphism (Table 12). The original formation temperatures of most SHMS deposits are thought to lie between 90 and 250 °C (cf. Table 3), corresponding to an expected range in PC 1* from roughly −1.7 to +3.0 (cf. Fig. 8a). Most of the reset values will clearly lie outside of this range. The consequence of this is that the mean value of PC 1* for SHMS deposits is shifted to a much lower value: -1.2 ± 0.3 in our dataset (cf. Fig. 7, corresponding to a mean GGIMFIS temperature of 270 ± 70 °C), virtually identical to the value for VHMS deposits. A primary signature closer to MVT deposits might have been expected from the range of original formation temperatures cited earlier, if no resetting had taken place. However, as illustrated in Table 4, about half of all SHMS deposits in our database were affected by greenschist facies or higher grade metamorphism, explaining this observation. While similar concerns apply to VHMS deposits, metamorphic overprinting should not generally result in as large a change to the mean value of PC 1* for these deposits, since their primary formation temperatures are significantly higher than those of SHMS deposits (cf. Table 3).

In conclusion, the most important effect of metamorphism will be a decrease in the value of PC 1, and therefore in Ga and Ge concentrations relative to the concentrations of Fe, Mn and, to a lesser degree, In. We note, however, that this does not necessarily mean that Ga and Ge will be expelled from the deposit, although increased mobility should occur. Instead, they might also partition into other minerals, e.g. the Cu–Fe–sulphides (e.g. Reiser et al., 2011; Belissont et al., 2015). Effects on other PCs were not studied in detail, reflecting both remaining uncertainties in their interpretation as well as the relative paucity of data on the chemistry of metamorphic sphalerite.

4.6. Relevance and applicability of results

The main contribution of this work lies in the establishment of a quantitative framework for the interpretation of sphalerite trace element data, expressed in terms of the definitions of the most relevant principal components (PCs 1, 2 and 3, perhaps 4). Another notable aspect is the demonstration that the first and second PCs correlate with fluid temperature and salinity, respectively, illustrating the utility of principal component analysis (PCA) to identify meaningful signals in a multivariate dataset without input of additional information.

While temperature control of Ga, Ge, Fe, Mn and, to a lesser degree, In concentrations in sphalerite had been suggested before (Graton and Harcourt, 1935; Oftedal, 1941; Stoiber, 1940; Warren and Thompson, 1945; Schroll, 1954, 1955; El Shazly et al., 1957), none of these previous authors backed up their claim with a detailed statistical analysis. Without testing against actual temperature data, as we have done here, their suggestions must have remained little more than speculation. The correlation of sphalerite composition (Ag, Cu, In and Ga concentrations) with the salinity of the ore-forming fluids represents an entirely new finding.

There are clear applications of our findings both in a scientific and economic context. First, the strong correlation between PC 1 and temperature provides (economic) geologists with a new tool for the study of ore-forming processes: the GGIMFis geothermometer. This appears to apply not only to hydrothermal but also to metamorphic processes. Second, the strong temperature control on the concentrations of Ga and Ge in sphalerite has clear implications for the identification of new potential sources of these elements.

In the context of mineral exploration, the observed correlation between In concentrations and the copper content of sphalerite and/or the incidence of chalcopyrite disease also has strong implications,

Table 12
Comparison of peak metamorphic and GGIMFIS temperatures for selected deposits.

Deposit, Country	Metamorphic facies	Type	T_{Peak} (°C)	PC1*	$T_{GGIMFIS}$ (°C)	No. of sph samples	Reference(s)
Vorta, Romania	Zeolite	VHMS	160 ± 40	1.2 ± 1.1	145 ± 115	1	*Chemistry*: Cook et al. (2009) *Temperature*: Ciobanu et al. (2001), Miron et al. (2012)
Eskay Creek, Canada	Subgreenschist	VHMS	225 ± 25	−0.1 ± 1.1	215 ± 115	1	*Chemistry*: Cook et al. (2009) *Temperature*: Sherlock et al. (1999), Meuzelaar (2015)
Utd. Verde, USA	Subgreenschist to lower greenschist	VHMS	250 ± 50	−0.5 ± 1.8	235 ± 150	2	*Chemistry*: Burnham (1959) *Temperature*: Anderson and Nash (1972), Vance and Condie (1987), Gustin (1990)
Mt. Isa, Australia	Lower greenschist	SHMS	310 ± 40	−1.8 ± 1.0	305 ± 110	2	*Chemistry*: Lockington et al. (2014) *Temperature*: Heinrich et al. (1989), Painter et al. (1999)
Røros, Norway	Lower greenschist to lower amphibolite	VHMS	400 ± 100	−1.9 ± 0.8	315 ± 100	3	*Chemistry*: Lockington et al. (2014) *Temperature*: Bjerkgård et al. (1999), Barrie et al. (2010)
Sulitjelma, Norway	Amphibolite	VHMS	540 ± 20	−1.7 ± 0.7	305 ± 90	5	*Chemistry*: Lockington et al. (2014) *Temperature*: Cook et al. (1993)
Mofjellet, Norway	Amphibolite	SHMS	550 ± 20	−1.5 ± 0.7	290 ± 90	3	*Chemistry*: Lockington et al. (2014) *Temperature*: Bjerkgård et al. (2001), Cook (2001)
Sauda, Norway	Amphibolite	VHMS	550 ± 100	−2.2 ± 1.1	325 ± 120	1	*Chemistry*: Cook et al. (2009) *Temperature*: Bingen et al. (2005)
Bleikvassli, Norway	Amphibolite	SHMS	570 ± 20	−2.0 ± 0.7	320 ± 95	4	*Chemistry*: Lockington et al. (2014) *Temperature*: Skauli et al. (1992), Cook (1993), Rosenberg et al. (1998)
Balmat, USA	Upper amphibolite	MVT	625 ± 25	−1.5 ± 0.2	290 ± 65	73	*Chemistry*: Doe (1960) *Temperature*: Whelan et al. (1984)
Marketorp, Sweden	Upper amphibolite	VHMS	650 ± 100	−2.4 ± 1.1	340 ± 120	1	*Chemistry*: Cook et al. (2009) *Temperature*: Sundblad (1994), Andersson (1997)
Zinkgruvan, Sweden	Upper amphibolite	VHMS	700 ± 100	−1.4 ± 1.1	280 ± 115	1	*Chemistry*: Cook et al. (2009) *Temperature*: Hedström et al. (1989), Sundblad (1994), Andersson (1997)
Broken Hill, Australia	Granulite	SHMS	775 ± 25	−1.7 ± 0.6	300 ± 80	2	*Chemistry*: Lockington et al. (2014) *Temperature*: Spry et al. (2008)

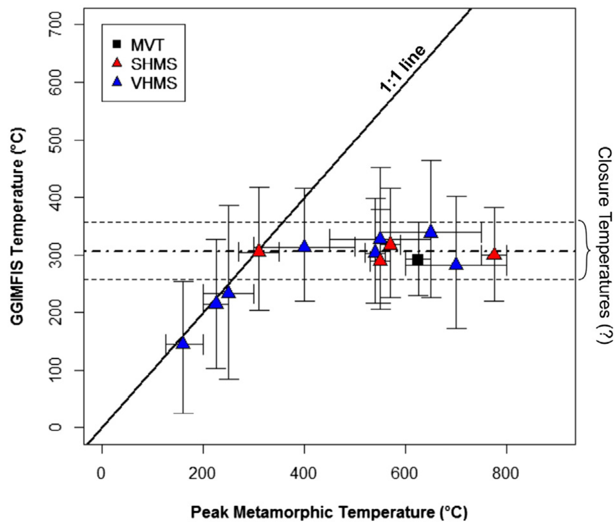


Fig. 12. Effect of metamorphism on sphalerite composition, as captured by changes in GGIMFis temperature. Note the clear departure from a 1:1 relationship between peak metamorphic temperature and GGIMFis temperature. Metamorphic grade data given in Table 12, see Electronic Annex A for complete plot data.

although this is not a new observation (Burnham, 1959; Johan, 1988; Cook et al., 2009, 2012). Namely, deposits where this is common would be expected to be more prospective for In. Once the exact control mechanism behind PC 2, which captures this relationship, can be determined, it will be useful in the identification of new sources of In.

Due to the complexity of the subject matter, our work necessarily leaves many open questions: What is the effect of sulphur fugacity, pH or redox-potential? To what extent and how are different fluid sources reflected in the composition of hydrothermal sphalerite? What might this tell us about the relevant ore-forming processes?

However, we also provide a framework in which these questions might be approached. Namely, research should focus on the identification of the relevant control mechanisms behind the still fully or partially unexplained PCs, particularly PC 2, PC 3 and PC 4. These represent, in decreasing order, the largest 'gaps' in our current understanding of the available data on sphalerite chemistry.

5. Summary and conclusions

To briefly summarise our main findings, we have shown that:

1) Significant differences exist in the compositions of sphalerite from different deposit types. These differences are mainly defined by the concentrations of Ga, Ge, Fe, Mn and, to a lesser extent In, and lie along a single dimension, the first principal component of the dataset, PC 1. Other PCs describe differences between individual deposits rather than deposit types.
2) PC 1 correlates strongly with the homogenisation temperature of fluid inclusions in sphalerite and associated minerals, probably reflecting temperature control on sphalerite composition. Therefore, the observed differences between deposit types can be explained entirely by differences in formation temperatures rather than fluid sources or other parameters. Concentrations of Fe, Mn and In increase, while those of Ga and Ge decrease with increasing formation temperature.
3) The correlation between PC 1 and temperature is strong enough to be used as a geothermometer (GGIMFis).
4) Metamorphic overprinting affects the value of PC 1 in accordance with its temperature dependence. However, no additional change is observed beyond a peak metamorphic temperature of ~310 ± 50 °C. This is probably due to the low closure temperature of the sphalerite system.
5) Independent of its temperature dependence, sphalerite composition also correlates with salinity. In this case, the second principal component, PC 2, is affected.
6) In addition to salinity, PC 2 correlates strongly with Cu concentrations. While this is highly speculative given the quality of our data, we suggest that this might reflect a general correlation between PC 2 and Cu activity in the ore-forming system. The correlation between Cu and In would therefore suggest that Cu-rich Pb–Zn systems are more prospective for In than Cu-poor ones.

The nature of the controlling factors behind the other PCs in our dataset still awaits identification and therefore poses interesting questions for future research. This should also focus on the extension of existing analytical datasets to include a more diverse range of elements. Particularly those elements present at reasonably high concentrations in sphalerite as substitutions for Zn (e.g. Hg, Tl) or S (e.g. Se) might provide additional geological insights.

Acknowledgements

The authors are greatly indebted to Prof Nigel Cook of the University of Adelaide, as well as two anonymous reviewers, for their comments and suggestions which helped to significantly improve this manuscript. Any remaining errors and inaccuracies are entirely our own. The handling editor, Dr. Cristiana Ciobanu, and editor-in-chief, Prof Franco Pirajno, are thanked for their patience. We would also like to thank Dr. Raimon Tolosana-Delgado and Prof K. Gerald van den Boogaart for many fruitful discussions about the statistical analysis of compositional data, as well as the help with the use of R. Last but not least, the authors would like to thank their respective host-institutions for the financial support without which this study would not have been possible.

Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.oregeorev.2015.12.017.

References

Abramson, B.S., 1981. The Mineralizing Fluids Responsible for Skarn and Ore Formation at the Continental Mine, Fierro, New Mexico, in Light of REE Analyses and Fluid Inclusion Studies (MSc thesis) New Mexico Institute of Mining and Technology, Sonora, New Mexico (143 pp.).

Agezo, F.L., 1995. Mineralogy, Alteration and Fluid Inclusion Study of the Lordsburg Mining District, Hidalgo County, New Mexico (MSc thesis) New Mexico Institute of Mining and Technology, Socorro, New Mexico (146 pp.).

Ahmad, S.N., Rose, A.W., 1980. Fluid inclusions in porphyry and skarn ore at Santa Rita, New Mexico. Econ. Geol. 75, 229–250.

Al-Bassam, K.S., Hak, J., Watkinson, D.H., 1982. Contribution to the origin of the Serguza lead–zinc–pyrite deposit, Northern Iraq. Mineral. Deposita 17, 133–149.

Alderton, D.H.M., Fallick, A.E., 2002. The nature and genesis of gold–silver–tellurium mineralization in the Metaliferi Mountains of Western Romania. Econ. Geol. 95, 495–516.

Anderson, C.A., Nash, J.T., 1972. Geology of the massive sulfide deposits at Jerome, Arizona – a reinterpretation. Econ. Geol. 67, 845–863.

Andersson, U.B., 1997. The late Svecofennian, high-grade contact and regional metamorphism in southwestern Bergslagen (central southern Sweden). Technical Report No. 970519. Swedish Geological Survey (29 pp.).

Appold, M.S., Wenz, Z.J., 2011. Composition of ore fluid inclusions from the Viburnum Trend, Southeast Missouri District, United States: implications for transport and precipitation mechanisms. Econ. Geol. 106, 55–78.

Axelsson, M.D., Rodushkin, I., 2001. Determination of major and trace elements in sphalerite using laser ablation double focusing sector field ICP-MS. J. Geochem. Explor. 72, 81–89.

Baker, T., Van Achterberg, E., Ryan, C.H., Lang, J.R., 2004. Composition and evolution of ore fluids in a magmatic hydrothermal skarn deposit. Geology 32, 117–120.

Barrie, C.D., Boyle, A.P., Cook, N.J., Prior, D.J., 2010. Pyrite deformation textures in the massive sulfide ore deposits of the Norwegian Caledonides. Tectonophysics 483, 269–286.

Barton, P.B., Bethke, P.M., 1987. Chalcopyrite disease in sphalerite: pathology and epidemiology. Am. Mineral. 72, 451–467.

Bauer, G., Ebert, A., Hesemann, J., von Kamp, H., Müller, D., Pietzner, H., Podufal, P., Scherp, A., Wellmer, F.-W., Arnold, O., Eckhardt, H., Herbst, F., Miederer, W., 1979. Die Blei-Zink-Erzlagerstätten von Ramsbeck und Umgebung. Monographien der deutschen Blei-Zink-Erzlagerstätten. Geol. Jahrb. 33.

Belissont, R., Boiron, M.-C., Luais, B., Cathelineau, M., 2014. LA-ICP-MS analyses of minor and trace elements and bulk Ge isotopes in zoned Ge-rich sphalerites from the Noailhac–Saint-Salvy deposit (France): insights into incorporation mechanisms and ore deposition processes. Geochim. Cosmochim. Acta 126, 518–540.

Belissont, R., Boiron, M.-C., Luais, B., Muchez, P., de Oliveira, D.P.S., Munoz, M., 2015. Germanium distribution and isotopic study in sulphides from MVT-related and VMS-remobilised ore deposits. In: André-Mayer, A.-S., et al. (Eds.), Mineral Resources in a Sustainable World, 13th Biennial SGA Meeting, Nancy, France, Proceedings, pp. 683–686.

Belissont, R., Munoz, M., Boiron, M.C., Luais, B., Mathon, O., 2016. Distribution and oxidation state of Ge, Cu and Fe in sphalerite by μ-XRF and K-edge μ-XANES: Insights into Ge incorporation, partitioning and isotopic fractionation. Geochim. Cosmochim. Acta (in press).

Bente, K., Doering, T., 1993. Solid-state diffusion in sphalerites: an experimental verification of the “chalcopyrite disease”. Eur. J. Mineral. 5, 465–478.

Bingen, B., Skår, Ø., Marker, M., Sigmond, E.M.O., Nordgulen, Ø., Ragnhildstveit, J., Mansfeld, J., Tucker, R.D., Liégeois, J.-P., 2005. Timing of continental building in the Sveconorwegian orogen, SW Scandinavia. Nor. J. Geol. 85, 87–116.

Bjerkgård, T., Sandstad, J.S., Sturt, B.A., 1999. Massive sulphide deposits in the South-Eastern Trundheim Region Caledonides, Norway: a review. In: Stanley, C.J., et al. (Eds.), Mineral Deposits: Processes to Processing, 5th Biennial SGA Meeting. A. A. Balkema, pp. 935–938.

Bjerkgård, T., Marker, M., Sandstad, J.S., Cook, N.J., Sordahl, T., 2001. Ore potential with emphasis on gold in the Mofjellet deposit, Rana, Nordland, Norway. Norges Geologiske Undersøkelse Technical Report 2001.050.

Bortnikov, N.S., Genkin, A.D., Dobrovol'skaya, M.G., Muravitskaya, G.N., Filimonova, A.A., 1991. The nature of chalcopyrite inclusions in sphalerite: exsolution, coprecipitation, or “disease”? Econ. Geol. 86, 1070–1082.

Bouch, J.E., Naden, J., Shepherd, T.J., McKervey, J.A., Young, B., Benham, A.J., Sloane, H.J., 2006. Direct evidence of fluid mixing in the formation of stratabound Pb–Zn–Ba–F mineralization in the Alston Block, North Pennine Orefield (England). Mineral. Deposita 41, 821–835.

Bresser, H.A., 1992. Origin of Base Metal Vein Mineralization in the Lawn Hill Mineral Field, Northwestern Queensland (BSc thesis) James Cook University (115 pp.).

Brewer, F.M., Cox, J.D., Morris, D.F.C., 1955. The occurrence of germanium in blende. Geochim. Cosmochim. Acta 8, 131–136.

Browne, R.H., 1979. On visual assessment of the significance of a mean difference. Biometrics 35, 657–665.

Burke, E.A.J., Kieft, C., 1980. Roquesite and Cu–In-bearing sphalerite from Långban, Bergslagen, Sweden. Can. Mineral. 18, 361–363.

Burnham, C.W., 1959. Metallogenic provinces of the Southwestern United States and Northern Mexico. N. M. Bur. Mines Min. Res. Bull. 65.

Bushnell, S.E., 1988. Mineralization at Cananea, Sonora, Mexico, and the paragenesis and zoning of breccia pipes in quartzofeldspathic rock. Econ. Geol. 83, 1760–1781.

Cabri, L.J., Campbell, J.L., LaFlamme, J.H.G., Leigh, R.G., Maxwell, J.A., 1985. Proton-microprobe analysis of trace elements in sulfides from massive-sulfide deposits. Can. Mineral. 23, 133–148.

Carpenter, A.B., Trout, M.L., Pickett, E.E., 1974. Preliminary report on the origin and chemical evolution of lead- and zinc-rich oil field brines in Central Mississippi. Econ. Geol. 69, 1191–1206.

Catchpole, H., Kouzmanov, K., Putlitz, B., Seo, J.H., Fontboté, L., 2015. Zoned base metal mineralization in a porphyry system: origin and evolution of mineralizing fluids in the Morococha District, Peru. Econ. Geol. 110, 39–71.

Cerny, I., Schroll, E., 1995. Heimische Vorräte an Spezialmetallen (Ga, In, Tl, Ge, Se und Cd) in Blei-Zink- und anderen Erzen. Arch. Lagerstättenforsch. Geol. Bundesanst. 18, 5–33.

Chakrabarti, A.K., 1967. On the trace element geochemistry of Zawar sulphides and its relation to metallogenesis. Can. Mineral. 9, 258–262.

Chambers, J.M., Freeny, A., Heiberger, R.M., 1992. Analysis of variance; designed experiments. In: Chambers, J.M., Hastie, T.J. (Eds.), Statistical Methods in S. Wadsworth and Brooks/Cole, Pacific Grove, California, pp. 145–194.

Cheng, M., Hu, X., Bao, Z., Bao, J., 2011. Geological features and metallogenesis of the Yutang Pb–Zn ore concentration belt in Hunan Province (in Chinese with English Abstract). Geol. Explor. 47, 251–260.

Ciobanu, C.L., Cook, N.J., Ivascanu, P.M., 2001. Ore deposits of the Vorta-Dealul Mare area, South Apuseni Mts., Romania: textures and a revised genetic model. Rom. J. Miner. Deposita 79, 46–47.

Clark, B.R., Kelly, W.C., 1973. Sulfide deformation studies: I. Experimental deformation of pyrrhotite and sphalerite to 2,000 bars and 500 °C. Econ. Geol. 68, 332–352.

Connolly, C.A., Walter, L.M., Baadsgaard, H., Longstaffe, F.J., 1990. Origin and evolution of formation waters, Alberta Basin, Western Canada sedimentary basin. I. Chemistry. Appl. Geochem. 5, 375–395.

Cook, N.J., 1993. Conditions of metamorphism estimated from alteration lithologies and ore at the Bleikvassli Zn–Pb–(Cu) deposit, Nordland, Norway. Nor. Geol. Tidsskr. 73, 226–233.

Cook, N.J., 2001. Ore mineralogical investigation of the Mofjell deposit (Mo i Rana, Nordland, Norway) with emphasis on gold and silver distribution. Norges geologiske undersøkelse Technical Report 2001.051.

Cook, N.J., Ciobanu, C.L., 2015. Mineral hosts for critical metals in hydrothermal ores. In: André-Mayer, A.-S., et al. (Eds.), Mineral Resources in a Sustainable World, 13th Biennial SGA Meeting, Nancy, France, Proceedings, pp. 707–710.

Cook, N.J., Halls, C., Boyle, A.P., 1993. Deformation and metamorphism of massive sulphides at Sulitjelma, Norway. Min. Mag. 57, 67–81.

Cook, N.J., Ciobanu, C.L., Pring, A., Skinner, W., Shimizu, M., Danyushevsky, L., Saini-Eidukat, B., Melcher, F., 2009. Trace and minor elements in sphalerite: a LA-ICPMS study. Geochim. Cosmochim. Acta 73, 4761–4791.

Cook, N.J., Sundblad, K., Valkama, M., Nygård, R., Ciobanu, C.L., Danyushevsky, L., 2011. Indium mineralization in A-type granites in southeastern Finland: insights into mineralogy and partitioning between coexisting minerals. Chem. Geol. 284, 62–73.

Cook, N.J., Ciobanu, C.L., Brugger, J., Etschmann, B., Howard, D.L., de Jonge, M.D., Ryan, C., Paterson, D., 2012. Determination of the oxidation state of Cu in substituted Cu–In–Fe-bearing sphalerite via μ-XANES spectroscopy. Am. Mineral. 97, 476–479.

Cook, N.J., Etschmann, B., Ciobanu, C.L., Geraki, K., Howard, D.L., Williams, T., Rae, N., Pring, A., Chen, G., Johannessen, B., Brugger, J., 2015. Distribution and substitution mechanism of Ge in Ge–(Fe)-bearing sphalerite. Minerals 5, 117–132.

Corbella, M., Avora, C., Cardellach, E., 2004. Hydrothermal mixing, carbonate dissolution and sulfide precipitation in Mississippi Valley-type deposits. Mineral. Deposita 39, 344–357.

Cox, D.P., Singer, D.A., 1986. Mineral deposit models. U.S. Geol. Surv. Bull. 1693.

Craig, J.R., Vaughan, D.J., 1994. Ore Microscopy and Ore Petrography. second ed. Wiley, New York.

Cureton, E.E., 1968. Unbiased estimation of the standard deviation. Am. Stat. 22, 22.

Dai, Y., Yu, X., Zhang, L., Cao, W., Zhu, Y., Li, C., 2014. Geology, isotopes and geochronology of the Caijiaping Pb–Zn deposit in the North Wuyi area, South China: implications for petrogenesis and metallogenesis. Ore Geol. Rev. 57, 116–131.

Debye, P., Hückel, E., 1923. Zur Theorie der Elektrolyte. I. Gefrierpunktserniedrigung und verwandte Erscheinungen. Phys. Z. 24, 185–206.

Development Core Team, R., 2004. A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria.

Doe, B.R., 1960. The Distribution and Composition of Sulfide Minerals at Balmat, New York (PhD thesis) California Institute of Technology, Pasadena, California (151 pp.).

Dunn, O.J., 1961. Multiple comparisons among means. J. Am. Stat. Assoc. 59, 52–64.

Dunnett, C.W., 1980. Pairwise multiple comparisons in the unequal variance case. J. Am. Stat. Assoc. 75, 796–800.

El Shazly, E.M., Webb, J.S., Williams, D., 1957. Trace elements in sphalerite, galena and associated minerals from the British Isles. Trans. Inst. Min. Metall. 66, 241–271.

Emsbo, P., 2000. Gold in Sedex deposits. Soc. Econ. Geol. Rev. 13, 427–437.

Erdmann, L., Graedel, T.E., 2011. Criticality of non-fuel minerals: a review of major approaches and analyses. Environ. Sci. Technol. 45, 7620–7630.

Erwood, R.J., Kesler, S.E., Cloke, P.L., 1979. Compositionally distinct, saline hydrothermal solutions, Naica Mine, Chihuahua, Mexico. Econ. Geol. 74, 95–108.

Ethier, V.G., Campbell, F.A., Both, R.A., Krouse, H.R., 1976. Geological setting of the Sullivan orebody and estimates of temperatures and pressure of metamorphism. Econ. Geol. 71, 1570–1588.

EU Commission, 2014. Report on critical raw materials for the EU. Report of the Ad-hoc Working Group on Defining Critical Raw Materials (Accessed on 1st March 2015, at http://ec.europa.eu/enterprise/policies/raw-materials/critical/index_en.htm).

Feiser, J., 1966. Nebenmetalle - Cadmium, Gallium, Germanium, Indium, Selen, Telur, Thallium, Wismut. Ferdinand Enke Verlag, Stuttgart.

Field, C.W., Moore, W.J., 1971. Sulfur isotope study of the “B” Limestone and Galena Fissure ore deposits of the U.S. Mine, Bingham mining district, Utah. Econ. Geol. 66, 48–62.

Fleischer, M., 1955. Minor elements in some sulfide minerals. In: Bateman, A.M. (Ed.), Economic Geology 50th Anniversary Volume. Economic Geology Publishing Co., Blacksburg, Va, pp. 970–1024.

Frenzel, M., Ketris, M.P., Gutzmer, J., 2014. On the geological availability of germanium. Mineral. Deposita 49, 471–486.

Frenzel, M., Tolosana-Delgado, R., Gutzmer, J., 2015. Assessing the supply potential of high-tech metals — a general method. Res. Policy 46, 45–58.

Frenzel, M., Ketris, M.P., Seifert, T., Gutzmer, J., 2016. On the current and future availability of gallium. Res. Policy 47, 38–50.

Fruth, I., Maucher, A., 1966. Spurenelemente und Schwefel-Isotope in Zinkblenden der Blei-Zink-Lagerstätte von Gorno. Mineral. Deposita 1, 238–250.

Fryklund, V.C., Fletcher, J.D., 1956. Geochemistry of sphalerite from the Star Mine, Coeur d'Alene district, Idaho. Econ. Geol. 51, 223–247.

Galaon, T., David, V., 2011. Deviation from van't Hoff dependence in RP-LC induced by tautomeric interconversion observed for four compounds. J. Sep. Sci. 34, 1423–1428.

Gardner, H.D., Hutcheon, I., 1985. Geochemistry, mineralogy and geology of the Jason Pb–Zn deposits, Macmillan Pass, Yukon, Canada. Econ. Geol. 80, 307–324.

Gasser, U., 1974. Zur Struktur und Geochemie der stratiformen Sulfidlagerstätte Meggen (Mitteldevon, Rheinisches Schiefergebirge). Geol. Rundsch. 63, 52–73.

George, L., Cook, N.J., Ciobanu, C.L., 2015a. Trace element partitioning between co-existing sphalerite, galena and chalcopyrite. In: André-Mayer, A.-S., et al. (Eds.), Mineral Resources in a Sustainable World, 13th Biennial SGA Meeting, Nancy, France, Proceedings, pp. 737–740.

George, L., Cook, N.J., Ciobanu, C.L., 2015b. Trace and minor elements in galena: a reconnaissance LA-ICP-MS study. Am. Mineral. 100, 548–569.

Goldschmidt, V.M., 1930. Über das Vorkommen des Germaniums in Steinkohlen und Steinkohlen-Produkten. Nachr. Ges. Wiss. Göttingen Math.-Phys. Kl. 1930, 398–401.

Graeser, S., 1969. Minor elements in sphalerite and galena from Binnatal (relations to coloration problems, isotopic composition etc.). Contrib. Mineral. Petrol. 24, 156–163.

Graton, L.C., Harcourt, G.A., 1935. Spectrographic evidence on origin of ores of Mississippi Valley type. Econ. Geol. 30, 800–824.

Guberman, D.E., 2015. Germanium. U.S. Geological Survey Mineral Commodity Summaries, January 2015. U.S. Geological Survey, Reston, Va, pp. 64–65.

Gustin, M.S., 1990. Stratigraphy and alteration of the host rocks, United Verde massive sulfide deposit, Jerome, Arizona. Econ. Geol. 85, 29–49.

Haditsch, J.G., Krainer, K., 1991. Jungalpidische Erzmineralisationen in der Phyllitgneiszone des Arlberggebietes (Tirol/Vorarlberg). Mitt. Österr. Geol. Ges. 84, 239–264.

Hall, W.E., Heyl, A.V., 1968. Distribution of minor elements in ore and host rock, Illinois–Kentucky fluorite district and Upper Mississippi Valley zinc–lead district. Econ. Geol. 63, 655–670.

Han, R., Liu, C., Huang, Z., Ma, D., Li, Y., Hu, B., Ma, G., Lei, L., 2004. Fluid inclusions of calcite and sources of ore-forming fluids in the Huize Zn–Pb–(Ag–Ge) district, Yunnan, China. Acta Geol. Sin. Engl. 78, 583–591.

Han, R.-S., Liu, C.-Q., Huang, Z.-L., Chen, J., Ma, D.-Y., Lei, L., Ma, G.-S., 2007. Geological features and origin of the Huize carbonate-hosted Zn–Pb–(Ag) district, Yunnan, South China. Ore Geol. Rev. 31, 360–383.

Hannington, M.D., de Ronde, C.E.J., Petersen, S., 2005. Sea-floor tectonics and submarine hydrothermal systems. In: Hedenquist, J.W., et al. (Eds.), Economic Geology 100th Anniversary Volume. Society of Economic Geologists, Inc., Littleton, Co., pp. 111–141.

Hawksworth, M.A., Meinert, L.D., 1990. Alteration and fluid inclusion study of the Groundhog vein system, Central Mining District, New Mexico. Econ. Geol. 85, 1825–1839.

Hedström, P., Simeonov, A., Malmström, L., 1989. The Zinkgruvan ore deposit, South-Central Sweden: a Proterozoic, proximal Zn–Pb–Ag deposit in distal volcanic facies. Econ. Geol. 84, 1235–1261.

Heinrich, C.A., Andrew, A.S., Wilkins, R.W.T., 1989. A fluid inclusion and stable isotope study of synmetamorphic copper ore formation at Mount Isa, Australia. Econ. Geol. 84, 529–550.

Huang, S., Zeng, Y., Jia, G., Chen, Y., 1987. On the genesis of Dabaoshan polymetallic deposit in Guangdong Province, China. Chin. J. Geochem. 6, 323–330.

Huichu, R., Renmin, H., Cox, D.P., 1991. Copper deposition by fluid mixing in deformed strata adjacent to a salt diapir, Dongchuan area, Yunnan province, China. Econ. Geol. 86, 1539–1545.

Huizenga, J.-M., Gutzmer, J., Banks, D., Greyling, L., 2006. The Paleoproterozoic carbonate-hosted Pering Zn–Pb deposit, South Africa. II: fluid inclusion, fluid chemistry and stable isotope constraints. Mineral. Deposita 40, 686–706.

Huston, D.L., Sie, S.H., Suter, G.F., Cooke, D.R., Both, R.A., 1995. Trace elements in sulfide minerals from Eastern Australian volcanic-hosted massive sulfide deposits: part I. Proton microprobe analyses of pyrite, chalcopyrite, and sphalerite, and part II. Selenium levels in pyrite: comparison with δ^{34}S values and implications for the source of sulfur in volcanogenic hydrothermal systems. Econ. Geol. 90, 1167–1196.

Hutchison, M.N., Scott, S.D., 1981. Sphalerite geobarometry in the Cu–Fe–Zn–S system. Econ. Geol. 76, 143–153.

Jaskula, B.W., 2015. Gallium. U.S. Geological Survey Mineral Commodity Summaries, January 2015. U.S. Geological Survey, Reston, Va, pp. 58–59.

Johan, Z., 1988. Indium and germanium in the structure of sphalerite: an example of coupled substitution with copper. Mineral. Petrol. 39, 211–229.

Jolly, J.L., Heyl, A.V., 1968. Mercury and other trace elements in sphalerite and wallrocks from Central Kentucky, Tennessee and Appalachian zinc districts. U.S. Geol. Surv. Bull. 1252-F.

Kappel, E., Schroll, E., 1982. Ablauf und Bildungstemperatur der Blei-Zink-Vererzungen von Bleiberg-Kreuth, Kärnten. Carinthia II 172, 49–62.

Kelley, K.D., Leach, D.L., Johnson, C.A., Clark, J.L., Fayek, M., Slack, J.F., Anderson, V.M., Ayuso, R.A., Ridley, W.I., 2004. Textural, compositional, and sulfur isotope variations of sulfide minerals in the Red Dog Zn–Pb–Ag deposits, Brooks Range, Alaska: implications for ore formation. Econ. Geol. 99, 1509–1532.

Kesler, S.E., Reich, M., Jean, M., 2007. Geochemistry of fluid inclusion brines from Earth's oldest Mississippi Valley-type (MVT) deposits, Transvaal Supergroup, South Africa. Chem. Geol. 237, 274–288.

Kharaka, Y.K., Maest, A.S., Carothers, W.W., Law, L.M., Lamothe, P.J., Fries, T.L., 1987. Geochemistry of metal-rich brines from central Mississippi salt dome basin, U.S.A. Appl. Geochem. 2, 543–561.

Klemm, L., Pettke, T., Graeser, S., Mullis, J., Kouzmanov, K., 2004. Fluid mixing as the cause of sulphide precipitation at Alpenbrunpass, Binn Valley, Central Alps. Schweiz. Miner. Petrog. 84, 189–212.

Koch, I., 2012. Analysis of multivariate and high-dimensional data theory and practice. Cambridge University Press, Cambridge, UK.

Koh, Y.-K., Choi, S.-G., So, C.-S., Choi, S.-H., Uchida, E., 1992. Application of arsenopyrite geothermometry and sphalerite geobarometry to the Taebaek Pn–Zn(—Ag) deposit Yeonhwa I mine, Republic of Korea. Mineral. Deposita 27, 58–65.

Kojima, S., Sugaki, A., 1984. Phase relations in the central portion of the Cu–Fe–Zn–S system between 800° and 500 °C. Mineral. J. 12, 15–28.

Kojima, S., Sugaki, A., 1985. Phase relations in the Cu–Fe–Zn–S system between 500° and 300 °C under hydrothermal conditions. Econ. Geol. 80, 158–171.

Kullerud, G., 1953. The FeS–ZnS system — a geological thermometer. Nor. Geol. Tidsskr. 32, 61–147.

Large, D., Walcher, E., 1999. The Rammelsberg massive sulphide Cu–Zn–Pb–Ba-Deposit, Germany: an example of sediment-hosted, massive sulphide mineralisation. Mineral. Deposita 34, 522–538.

Lau, M.K., 2013. DTK: Dunnett–Tukey–Kramer Pairwise Multiple Comparison Test Adjusted for Unequal Variances and Unequal Sample Sizes. (Accessed on 3rd March 2015 at) http://cran.r-project.org/web/packages/DTK/index.html.

Leach, D.L., Nelson, R.C., Williams, D., 1975. Fluid inclusion studies in the Northern Arkansas zinc district. Econ. Geol. 70, 1084–1091.

Leach, D.L., Landis, G.P., Hofstra, A.H., 1988. Metamorphic origin of the Coeur d'Alene base- and precious-metal veins in the Belt basin, Idaho and Montana. Geology 16, 122–125.

Leach, D.L., Marsh, E., Emsbo, P., Rombach, C., Kelley, K.D., Reynolds, J., Anthony, M., 2004. Nature of hydrothermal fluids at the shale-hosted Red Dog Zn–Pb–Ag deposits, Brooks Range, Alaska. Econ. Geol. 99, 1449–1480.

Leach, D.L., Sangster, D.F., Kelley, K.D., Large, R.R., Garven, G., Allen, C.R., Gutzmer, J., Walters, S., 2005. Sediment-hosted lead–zinc deposits: a global perspective. In: Hedenquist, J.W., et al. (Eds.), Economic Geology 100th Anniversary Volume. Society of Economic Geologists, Inc., Littleton, Co., pp. 561–607.

Lecoq de Boisbeaudran, P.-E., 1875. Caractères chimiques et spectroscopiques d'un nouveau métal, le Gallium, découvert dans une blende de la mine de Pierrefitte, vallée d' Argelès (Pyrénées). C. R. Hebd. Acad. Sci. 81, 493–495.

Li, J., Kusky, T.M., 2007. World's largest known Precambrian fossil black smoker chimneys and associated microbial vent communities, North China: implications for early life. Gondwana Res. 12, 84–100.

Li, B., Gu, X., Han, R., Wen, S., 2011. Trace element geochemistry of sphalerite in Huize lead–zinc deposit, northeast Yunnan, China. International Conference on Multimedia Technology (ICMT), Hangzhou, p. 2011 http://dx.doi.org/10.1109/ICMT.2011.6002858.

Lockington, J.A., Cook, N.J., Ciobanu, C.L., 2014. Trace and minor elements in sphalerite from metamorphosed sulphide deposits. Mineral. Petrol. 108, 873–890.

Lovering, T.S., Tweto, O., Lovering, T.G., 1978. Ore deposits of the Gilman District, Eagle County, Colorado. U. S. Geol. Surv. Prof. Paper 1017.

Lu, H., 1983. Fluid inclusion study of Fankou Pb–Zn ore deposit, Fankou, Guangdong, China. Chin. J. Geochem. 2, 45–57.

Lusk, J., Krouse, H.R., 1997. Comparative stable isotope and temperature investigation of minerals and associated fluids in two regionally metamorphosed (Kuroko-type) volcanogenic massive sulfide deposits. Chem. Geol. 143, 231–253.

Lusk, J., Scott, S.D., Ford, C.E., 1993. Phase relations in the Fe–Zn–S system to 5 kbars and temperatures between 325° and 150 °C. Econ. Geol. 88, 1880–1903.

Manske, S., Ullrich, T., Reynolds, T.J., O'Connor, G., 2006. Vein sets and hydrothermal alteration in the Cetate-Carnic area, Rosia Montana District, Romania. Rom. J. Miner. Deposita 81, 122–125.

Mariko, T., Kawada, M., Miura, M., Ono, S., 1996. Ore formation processes of the Mozumi skarn-type Pb–Zn–Ag deposit in the Kamioka Mine, Gifu Prefecture, Central Japan — a mineral chemistry and fluid inclusion study. Resour. Geol. 46, 337–354.

McIntire, W.L., 1963. Trace element partition coefficients — a review of theory and applications to geology. Geochim. Cosmochim. Acta 27, 1209–1264.

McKnight, E.T., Fisher, R.P., 1970. Geology and ore deposits of the Picher field. Oklahoma and Kansas U. S. Geol. Surv. Prof. Paper 588.

McLemore, V.T., 1993. Geology and geochemistry of the mineralization and alteration in the Steeple Rock District, Grant County, New Mexico and Greenlee County, Arizona. New Mexico Bureau of Mines and Mineral Resources Open-file Report 397.

McLimans, R.K., 1977. Geochemical, fluid inclusion and stable isotope studies of the Upper Mississippi Valley zinc–lead district, southwest Wisconsin (PhD thesis) Pennsylvania State University, State College (175 pp.).

Meinert, L.D., 1982. Skarn, manto, and breccia pipe formation in the sedimentary rocks of the Cananea mining district, Sonora, Mexico. Econ. Geol. 77, 919–949.

Meinert, L.D., 1987. Skarn zonation and fluid evolution in the Groundhog Mine, Central mining district, New Mexico. Econ. Geol. 82, 523–545.

Meinert, L.D., Dipple, G.M., Nicolescu, S., 2005. World skarn deposits. In: Hedenquist, J.W., et al. (Eds.), Economic Geology 100th Anniversary Volume. Society of Economic Geologists, Inc., Littleton, Co., pp. 299–336.

Meuzelaar, T., 2015. Hydrothermal Alteration of Carbonaceous Mudstones Hosting the Eskay Creek Au Deposit, British Columbia (MSc thesis) Colorado School of Mines, Golden, Colorado (178 pp.).

Miron, G.D., Neuhoff, P.S., Amthauer, G., 2012. Low-temperature hydrothermal metamorphic mineralization of island-arc volcanics, South Apuseni Mountains, Romania. Clay Clay Miner. 60, 1–17.

Moenke-Blankenburg, L., Kammel, J., Schumann, T., 1994. Microanalysis by Laser Ablation-Inductively Coupled Plasma-Atomic Emission Spectrometry in comparison to spark ablation of certified and found minerals galena, sphalerite, and pyrite. Microchem. J. 50, 374–384.

Möller, P., 1987. Correlation of homogenization temperatures of accessory minerals from sphalerite-bearing deposits and Ga/Ge model temperatures. Chem. Geol. 61, 153–159.

More, A.P., Vaughan, D.J., Ashworth, J.R., 1991. Banded sphalerite from the North Pennine Orefield. Mineral. Mag. 55, 409–416.

Morris, D.F.C., Brewer, F.M., 1954. The occurrence of gallium in blende. Geochim. Cosmochim. Acta 5, 134–141.

Mortimer, C.E., Müller, U., 2001. Chemie. Das Basiswissen der Chemie. eighth ed. Thieme, Stuttgart.

Munoz, M., Boyce, A.J., Courjault-Rade, P., Fallick, A.E., Tollon, F., 1994. Multi-stage fluid incursion in the Palaeozoic basement-hosted Saint-Salvy ore deposit (NW Montagne Noire, southern France). Appl. Geochem. 9, 609–626.

Murakami, H., Ishihara, S., 2013. Trace elements of Indium-bearing sphalerite from tin-polymetallic deposits in Bolivia, China and Japan: a femto-second LA-ICP-MS study. Ore Geol. Rev. 53, 223–243.

Oftedal, I., 1941. Untersuchungen über die Nebenbestandteile von Erzmineralen norwegischer zinkblende-führender Vorkommen. Norsk. Vidensk. Akad. Skr. Mat.-Nat. Kl. 8, 1–103.

Olade, M.A., Morton, R.D., 1985. Origin of lead–zinc mineralization in the southern Benue Trough, Nigeria — fluid inclusion and trace element studies. Mineral. Deposita 20, 76–80.

Painter, M.G.M., Golding, S.D., Hannan, K.W., Neudert, M.K., 1999. Sedimentologic, petrographic, and sulfur isotope constraints on fine-grained pyrite formation at Mount Isa mine and environs, Northwest Queensland, Australia. Econ. Geol. 94, 883–912.

Palero, F.J., Both, R.A., Arribas, A., Boyce, A.J., Mangas, J., Martin-Izard, A., 2003. Geology and metallogenic evolution of the polymetallic deposits of the Alcudia Valley mineral field, Eastern Sierra Morena, Spain. Econ. Geol. 98, 577–605.

Palero-Fernández, F.J., Martín-Izard, A., 2005. Trace element contents in galena and sphalerite from ore deposits of the Alcudia Valley mineral field (Eastern Sierra Morena, Spain). J. Geochem. Explor. 86, 1–25.

Papish, J., Brewer, F.M., Holt, D.A., 1927. Germanium XXV. Arc spectrographic detection and estimation of germanium. Occurrence of germanium in certain tin minerals. Enargite as a possible source of germanium. J. Am. Chem. Soc. 49, 3028–3033.

Pattrick, R.A.D., Bowell, R.J., 1991. The genesis of the West Shropshire Orefield: evidence from fluid inclusions, sphalerite chemistry, and sulphur isotopic ratios. Geol. J. 26, 101–115.

Payton, M.E., Greenstone, M.H., Schenker, N., 2003. Overlapping confidence intervals or standard errors: what do they mean in terms of statistical significance. J. Insect Sci. 3, 34.
Penney, S.R., Allen, R.M., Harrison, S., Lees, T.C., Murphy, F.C., Norman, A.R., 2004. The global distribution of zinc mineralisation, an analysis based on a new zinc deposits database. Appl. Earth Sci. 113, 171–182.
Polito, P.A., Kyser, T.K., Golding, S.D., Southgate, P.N., 2006. Zinc deposits and related mineralization of the Burketown mineral field, including the world-class Century deposit, Northern Australia: fluid inclusion and stable isotope evidence for basin fluid sources. Econ. Geol. 101, 1251–1273.
Prantner, B., 2011. Visualization of Imputed Values Using the R-package VIM. (Accessed on 25th February 2015 at) http://cran.r-project.org/web/packages/VIMGUI/vignettes/VIM-Imputation.pdf.
Qian, Z., 1987. Trace elements in galena and sphalerite and their geochemical significance in distinguishing the genetic types of Pb–Zn ore deposits. Chin. J. Geochem. 6, 177–190.
Ramdohr, P., 1975. Die Erzminerale und ihre Verwachsungen. fourth ed. Akademie-Verlag, Berlin.
Rasmy, M., 1981. Trace-elements content of galenas and associated minerals in some Miocene lead–zinc deposits near Red Sea coast, Egypt. Geol. Rundsch. 70, 874–881.
Reich, F., Richter, T., 1863. Über das Indium. J. Prakt. Chem. 90, 172–176.
Reiser, F.K.M., Rosa, D.R.N., Pinto, A.M.M., Carvalho, J.R.S., Matos, J.X., Guimaraes, F.M.G., Alves, L.C., de Oliveira, D.P.S., 2011. Mineralogy and geochemistry of tin- and germanium-bearing copper ore, Barrigao re-mobilized vein deposit, Iberian Pyrite Belt, Portugal. Int. Geol. Rev. 53, 1212–1238.
Richardson, C.K., Pinckney, D.M., 1984. The chemical and thermal evolution of the fluids in the Cave-in-Rock fluorspar distict, Illinois: mineralogy, paragenesis, and fluid inclusions. Econ. Geol. 79, 1833–1856.
Roedder, E., 1971. Fluid inclusion studies on the porphyry-type ore deposits at Bingham, Utah, Butte, Montana, and Climax, Colorado. Econ. Geol. 66, 98–120.
Rose, A.W., 1967. Trace elements in sulfide minerals from the Central district, New Mexico and the Bingham district, Utah. Geochim. Cosmochim. Acta 31, 547–585.
Rosenberg, J.L., Spry, P.G., Jacobson, C.E., Cook, N.J., Vokes, F.M., 1998. Thermobarometry of the Bleikvassli Zn–Pb–(Cu) deposit, Nordland, Norway. Mineral. Deposita 34, 19–34.
Rowan, E.L., Goldhaber, M.B., 1996. Fluid inclusions and biomarkers in the Upper Mississippi Valley zinc–lead district — implications for the fluid-flow and thermal history of the Illinois Basion. U.S. Geol. Surv. Bull., 2094-F
Samson, I.M., Williams-Jones, A.E., Ault, K.M., Gagnon, J.E., Fryer, B.J., 2008. Source of fluids forming distal Zn–Pb–Ag skarns: evidence from laser ablation-inductively coupled plasma-mass sepctrometry analysis of fluid inclusions from El Mochito, Honduras. Geology 36, 947–950.
Sangster, D.F., Nowlan, G.S., McCracken, A.D., 1994. Thermal comparison of Mississippi Valley-type lead zinc deposits and their host rocks using fluid inclusion and conodont color alteration index data. Econ. Geol. 89, 493–514.
Sawkins, F.J., 1964. Lead–zinc ore deposition in the light of fluid inclusion studies, Providencia Mine, Zacatecas, Mexico. Econ. Geol. 59, 883–919.
Schaefer, M.O., Gutzmer, J., Beukes, N.J., Greyling, L.N., 2004. Mineral chemistry of sphalerite and galena from Pb–Zn mineralization hosted by the Transvaal Supergroup in Griqualand West, South Africa. S. Afr. J. Geol. 107, 341–354.
Schopfhauser, D., Templ, M., Alfons, A., Kowarik, A., Pantner, B., 2014. VIMGUI: Visualization and Imputation of Missing Values. (Accessed on 10th March 2015 at) http://cran.r-project.org/web/packages/VIMGUI/index.html.
Schroll, E., 1954. Ein Beitrag zur geochemischen Analyse ostalpiner Blei-Zink-Erze. Mitt. Österr. Miner. Ges. Sonderbd. 3, 1–85.
Schroll, E., 1955. Über das Vorkommen einiger Spurenmetalle in Blei-Zink-Erzen der ostalpinen Metallprovinz. Tscher. Miner. Petrogr. 5, 183–208.
Scott, S.D., Kissin, S.A., 1973. Sphalerite composition in the Zn–Fe–S system below 300 °C. Econ. Geol. 68, 475–479.
Sherlock, R.L., Roth, T., Spooner, E.T.C., Bray, C.J., 1999. Origin of the Eskay Creek precious metal-rich volcanogenic massive sulfide deposit: fluid inclusion and stable isotope evidence. Econ. Geol. 94, 803–824.
Shikazono, N., 1974. Physico-chemical properties of ore-forming solution responsible for the formation of Toyoha Pb–Zn deposits, Hokkaido, Japan. Geochem. J. 8, 37–46.
Simmons, S.F., White, N.C., John, D.A., 2005. Geological characteristics of epithermal precious and base metal deposits. In: Hedenquist, J.W., et al. (Eds.), Economic Geology 100th Anniversary Volume. Society of Economic Geologists, Inc., Littleton, Co., pp. 485–522.
Skauli, H., Boyce, A.J., Fallick, A.E., 1992. A sulphur isotopic study of the Bleikvassli Zn–Pb–Cu deposit, Nordland, northern Norway. Mineral. Deposita 27, 284–292.
So, C.-S., 1977. Spurenelement-Paragenesen in Zinkbklenden verschiedener Pb-Zn-Vorkommen in Korea. J. Mineral. Petrol. Sci. 8, 439–455.
So, C.-S., Yun, S.-T., 1992. Geochemistry and genesis of hydrothermal Au–Ag–Pb–Zn deposits in the Hwanggangri mineralized district, Republic of Korea. Econ. Geol. 87, 2056–2084.
So, C.-S., Park, M.-E., Shelton, K.L., Seidemann, D.E., 1984. Geology and geochemistry of the Sambo Pb–Zn deposit, Republic of Korea. Econ. Geol. 79, 656–670.
Sober, E., 1994. Let's razor Occam's Razor. In: Knowles, D. (Ed.), Explanation and Its Limits. Cambridge University Press, Cambridge.
Sperling, H., Lange, J., von Scotti, H.-H., 1973. Die Blei-Zink-Erzgänge des Oberharzes (Die Erzgänge des Erzbergwerkes Grund (Silbemaaler Gangzug, Bergwerksglücker Gang und Laubhütter Gang)). Monographien der Deutschen Blei-Zink-Erzlagerstätten. Geol. Jahrb. D2.
Spry, P.G., Fuhrmann, G.D., 1994. Additional fluid inclusion data for the Illinois–Kentucky fluorspar district: evidence for the lack of regional thermal gradient. Econ. Geol. 89, 288–306.
Spry, P.G., Plimer, I.R., Teale, G.S., 2008. Did the giant Broken Hill (Australia) Zn–Pb–Ag deposit melt? Ore Geol. Rev. 34, 223–241.
Stanimirova, I., Daszykowski, M., Walczak, B., 2007. Dealing with missing values and outliers in principal component analysis. Talanta 72, 172–178.
Stanley, C.J., Vaughan, D.J., 1982. Copper, lead, zinc and cobalt mineralization in the English Lake District: classification, conditions of formation and genesis. J. Geol. Soc. Lond. 139, 569–579.
Stevens, S.S., 1946. On the theory of scales of measurement. Science 103, 677–680.
Stoffell, B., Wilkinson, J.J., Jeffries, T.E., 2004. Metal transport and deposition in hydrothermal veins revealed by 213 nm UV laser ablation microanalysis of single fluid inclusions. Am. J. Sci. 304, 533–557.
Stoffell, B., Appold, M.S., Wilkinson, J.J., McClean, N.A., Jeffries, T.E., 2008. Geochemistry and evolution of Mississippi Valley-Type mineralizing brines from the Tri-State and Northern Arkansas districts determined by LA-ICP-MS microanalysis of fluid inclusions. Econ. Geol. 103, 1411–1435.
Stoiber, R.E., 1940. Minor elements in sphalerite. Econ. Geol. 35, 501–519.
Stueber, A.M., Walter, L.M., 1991. Origin and chemical evolution of formation waters from Silurian–Devonian strata in the Illinois basin, USA. Geochim. Cosmochim. Ac. 55, 309–325.
Sundblad, K., 1994. A genetic reinterpretation of the Falun and Åmmeberg ore types, Bergslagen, Sweden. Mineral. Deposita 29, 170–179.
Templ, M., Kowarik, A., Filzmoser, P., 2011. Iterative regression imputation using standard and robust methods. Comput. Stat. Data Anal. 55, 2793–2806.
Tolcin, A. C., 2015. Indium, in: U.S. Geological Survey Mineral Commodity Summaries, January 2015. U.S. Geological Survey, Reston, Va. pp. 74–75.´
Turner, D.R., Bowman, J.R., 1993. Origin and evolution of skarn fluids, Empire zinc skarns, Central Mining District, New Mexico, USA. Appl. Geochem. 8, 9–36.
van den Boogaart, K.G., Tolosana-Delgado, R., 2013. Analyzing Compositional Data With R. first ed. Springer, Heidelberg-New York-Dodrecht-London.
Vance, R.K., Condie, K.C., 1987. Geochemistry of footwall alteration associated with the Early Proterozoic United Verde massive sulfide deposit, Jerome, Arizona. Econ. Geol. 82, 571–586.
Vavtar, F., 1988. Die Erzanreicherungen im Nordtiroler Stubai-, Ötztal- und Silvrettakristallin. Arch. Lagerstättenforsch. Geol. Bundesanst. 9, 103–153.
Viets, J.G., Hopkins, R.T., Miller, B.M., 1992. Variations in minor and trace metals in sphalerite from Mississippi valley-type deposits of the Ozark region; genetic implications. Econ. Geol. 87, 1897–1905.
Vikre, P.G., 1998. Intrusion-related, polymetallic carbonate replacement deposits in the Eureka District, Eureka County, Nevada. Nev. Bur. Min. Geol. Bull. 110.
Vivallo, W., Rickard, D., 1990. Genesis of an Early Proterozoic zinc deposit in high-grade metamorphic terrane, Saxberget, Central Sweden. Econ. Geol. 80, 17–32.
Wallier, S., Rey, R., Kouzmanov, K., Pettke, T., Heinrich, C.A., et al., 2006. Magmatic fluids in the breccia-hosted epithermal Au–Ag deposit of Rosia Montana, Romania. Econ. Geol. 101, 923–954.
Warren, H.V., Thompson, R.M., 1945. Sphalerites from western Canada. Econ. Geol. 40, 309–335.
Watling, R.J., 1976. Trace element distribution in primary sulphide minerals from the Keel Prospect, County Longford. Proc. Roy. Irish Acad. 76, 241–261.
Weber, L., 1990. Die Blei-Zinkerzlagerstätten des Grazer Paläozoikums und ihr geologischer Rahmen. Arch. Lagerstättenforsch. Geol. Bundesanst. 12, 1–289.
Whelan, J.F., Rye, R.O., DeLorraine, W., 1984. The Balmat–Edwards zinc–lead deposits — synsedimentary ore from Mississippi Valley-type fluids. Econ. Geol. 79, 239–265.
Wilkinson, J.J., 2001. Fluid inclusions in hydrothermal ore deposits. Lithos 55, 229–272.
Wilkinson, J.J., 2010. A review of fluid inclusion constraints on mineralization in the Irish ore field and implications for the genesis of sediment-hosted Zn–Pb deposits. Econ. Geol. 105, 417–442.
Winderbaum, L., Ciobanu, C.L., Cook, N.J., Paul, M., Metcalfe, A., Gilbert, S., 2012. Multivariate analysis of an LA-ICP-MS trace element dataset for pyrite. Math. Geosci. 44, 823–842.
Winkler, C., 1886. Germanium, Ge, ein neues, nichtmetallisches Element. Ber. Dtsch. Chem. Ges. 19, 210–211.
Wood, S.A., Samson, I.M., 2006. The aqueous geochemistry of gallium, germanium, indium and scandium. Ore Geol. Rev. 28, 57–102.
Wright, S., 1921. Correlation and causation. J. Agric. Res. 20, 557–585.
Wu, J., 1985. Geological characteristics of the Xitieshan massive sulfide deposit, Qinghai province (in Chinese with English abstract). Mineral. Deposita 4, 1–12.
Xia, X., Yuan, J., Zhao, Y., Yan, F., 2003. Geology of the polymetallic pyrite deposits along the northern margin of north China platform with a focus on regional division of prospective areas (in Chinese with English abstract). Geol. Chem. Min. 25, 129–144.
Xia, Q.-L., Chen, Y., Lu, Y.-X., Jiang, C.-X., Liu, H.-G., Lu, Z.-C., 2005. Geochemistry, fluid inclusion, and stable isotope studies of Luziyuan Pb–Zn deposit in Yunnan Province, Southwestern China (in Chinese with English abstract). J. China Univ. Geosci. 30, 177–186.
Xue, C., Zeng, R., Liu, S., Chi, G., Qing, H., Chen, Y., Yang, J., Wang, D., 2007. Geologic, fluid inclusion and isotopic characteristics of the Jinding Zn–Pb deposit, western Yunnan, South China: a review. Ore Geol. Rev. 31, 337–359.
Xuexin, S., 1984. Minor elements and ore genesis of the Fankou lead–zinc deposit, China. Mineral. Deposita 19, 95–104.
Yajima, J., Ohta, E., 1979. Two-stage mineralization and formation process of the Toyoha deposits, Hokkaido, Japan, Min. Geol. 29, 291–306.
Yang, D.Y., Utsugi, Y., Mariko, T., 1993. Fluid inclusion study on magnesian Fe skarn-type deposit of the Shinyemi Mine, Republic of Korea. Resour. Geol. 43, 11–22.
Yardley, B.W.D., 2005. Metal concentrations in crustal fluids and their relationship to ore formation. Econ. Geol. 100, 613–632.
Yardley, B.W.D., Bodnar, R.J., 2014. Fluids in the continental crust. Geochem. Perspect. 3, 1–127.

Yardley, B.W.D., Cleverley, J.S., 2013. The role of metamorphic fluids in the formation of ore deposits. In: Jenkin, G.R.T., et al. (Eds.), Ore Deposits in an Evolving Earth. Geological Society, London, Special Publications 393, pp. 117–134.

Ye, L., Cook, N.J., Ciobanu, C.L., Yuping, L., Qian, Z., Tiegeng, L., Wei, G., Yulon, Y., Danyushevskiy, L., 2011. Trace and minor elements in sphalerite from base metal deposits in South China: a LA-ICPMS study. Ore Geol. Rev. 39, 188–217.

Ye, L., Cook, N.J., Liu, T., Ciobanu, C.L., Gao, W., Yang, Y., 2012. The Niujiaotang Cd-rich zinc deposit, Duyun, Guizhou province, southwest China: ore genesis and mechanisms of cadmium concentration. Mineral. Deposita 47, 683–700.

Yi, W., Halliday, A., Lee, D.-C., Christensen, J.N., 1995. Indium and tin in basalts, sulfides, and the mantle. Geochim. Cosmochim. Acta 59, 5081–5090.

Yun, S., Einaudi, M.T., 1982. Zinc–lead skarns of the Yeonhwa-Ulchin district, South Korea. Econ. Geol. 77, 1013–1032.

Zaw, K., Gemmell, J.B., Large, R.R., Mernagh, T.P., Ryan, C.G., 1996. Evolution and source of ore fluids in the stringer system, Hellyer VHMS deposit, Tasmania, Australia: evidence from fluid inclusion microthermometry and geochemistry. Ore Geol. Rev. 10, 251–278.

Zaw, K., Hunns, S.R., Large, R.R., Gemmell, J.B., Ryan, C.G., Mernagh, T.P., 2003. Microthermometry and chemical composition of fluid inclusions from the Mt Chalmers volcanic-hosted massive sulfide deposits, central Queensland, Australia: implications for ore genesis. Chem. Geol. 194, 225–244.

Zeeh, S., Bechstädt, T., 1994. Carbonate-hosted Pb–Zn mineralization at Bleiberg-Kreuth (Austria): compilation of data and new aspects. In: Boni, M., Fontboté, L. (Eds.), Sediment-hosted Zn–Pb Ores. Springer, Heidelberg, pp. 271–296.

Zeeh, S., Kuhlemann, J., Bechstädt, T., 1998. The classical Pb–Zn deposits of the Eastern Alps (Austria/Slovenia) revisited: MVT deposits resulting from gravity driven fluid flow in the Alpine realm. Geologija 41, 257–273.

Zhou, J., Huang, Z., Zhou, G., Li, X., Ding, W., Bao, G., 2011. Trace elements and rare earth elements of sulfide minerals in the Tianqiao Pb–Zn ore deposit, Guizhou Province, China. Acta Geol. Sin. 85, 189–199.

Update

Ore Geology Reviews
Volume 81, Issue P1, March 2017, Page 400

DOI: https://doi.org/10.1016/j.oregeorev.2016.10.009

Ore Geology Reviews 81 (2017) 400

Contents lists available at ScienceDirect

Ore Geology Reviews

journal homepage: www.elsevier.com/locate/oregeorev

Corrigendum

Corrigendum to “Gallium, germanium, indium and other trace and minor elements in sphalerite as a function of deposit type – A meta-analysis” [Ore Geol. Rev. 76 (2016) 52–78]

Max Frenzel [a,*], Tamino Hirsch [a], Jens Gutzmer [a,b]

[a] Helmholtz-Zentrum Dresden-Rossendorf, Helmholtz Institute Freiberg for Resource Technology, Halsbrücker Str. 34, 09599 Freiberg, Germany
[b] Institute for Mineralogy, University of Technology Bergakademie Freiberg, Brennhausgasse 14, 09599 Freiberg, Germany

The abstract of the original version of this article contained an unfortunate mistake. In particular, the formula for the GGIMFis geothermometer given in the abstract was erroneous, missing a minus sign before the factor 54.4 with which the value of PC1* is multiplied. The correct formula is:

$$T(\,^{\circ}\mathrm{C}) = -(54.4 \pm 7.3) \cdot \mathrm{PC1} * +(208 \pm 10)$$

and is also given in Fig. 11 of the original article.

We thank Prof. Frank Melcher of the Montanuniversität Leoben for alerting us to this error.

DOI of original article: http://dx.doi.org/10.1016/j.oregeorev.2015.12.017.
* Corresponding author.
E-mail address: m.frenzel@hzdr.de (M. Frenzel).

http://dx.doi.org/10.1016/j.oregeorev.2016.10.009